

The ES–Fable inverse correspondence in the original weighted conductor

Split-Zero programme: coordinate calculation

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This calculation responds to equation (11) of the supplied audit: the inverse-exterior bound for the matrix in WCF6 [3]. It constructs the marked map from the four ES–Fable locales, follows it through the original quartet and degree-eight conductor, and determines where the receiving finite linear map can lose rank. All four labels are retained. The conclusions concern this precise map. They neither assert RH nor classify all nonlinear maps in the programme. The original three-dimensional polynomial example was announced by Levent Alpöge with credit to Fable; Tao’s exposition [6] reproduces its formula. The additional four-Time constructions used here are credited to the ES programme source [2], rather than attributed to that original three-point fibre.

The continuation solves the entire inverse problem for the displayed four-dimensional map: every finite fibre, its complete omitted image, its degree-eight inverse domain and its complete nonproper-value locus. It gives the eight signed states over the literal arithmetic quartet, their common limiting omitted target, and their escape in a fixed original Gamma norm. For the earlier seven-point target it evaluates the complete singular spectrum, all exterior inverse asymptotes, and the exact volume distortion of explicitly transported coordinate flags. The original conductor intertwines these maps throughout; its coefficients are never chosen to fit the new polynomial.

The signed-state continuation gives a nonzero exact determinant for the four sign differences on the literal quartet, the full rank-three mean map and its explicit kernel line, and the entire eight-state evaluation kernel. Retaining labels and evaluation gives an invertible total map through two copies of the original conductor. Its signed-root symmetry has order 192; all monodromy generators, its deck transformations, the complete mixing block and its ranks are calculated explicitly.

1 Four points, including the coordinates hidden by local symmetry

The source [2], equations `four-Time-quarter-locales`, `times-weight-and-unit-vectors`, and `four-Time-quarter-coordinate-matrix`, defines

$$\mathcal{T} = (M, N, T, E), \quad V = \{v \in \mathbb{R}^4 : \mathbf{1}^T v = 0\}, \quad \mu_a = \frac{2}{\sqrt{3}}(e_a - \tfrac{1}{4}\mathbf{1}), \quad \mathfrak{q}_a = -\tfrac{1}{3}\mu_a. \quad (\text{FC1})$$

Here e_a are the four separate coordinate vectors; no two are identified. Consequently

$$\langle \mathfrak{q}_a, \mathfrak{q}_b \rangle = \begin{cases} 1/9 & a = b, \\ -1/27 & a \neq b, \end{cases} \quad Q_{ab} := \tfrac{3}{4}\langle \mathfrak{q}_b, \mu_a \rangle = \tfrac{1}{12} - \tfrac{1}{3}\delta_{ab}. \quad (\text{FC2})$$

The matrix Q has four distinct columns, although each column has its own local altitude $-1/4$. Direct multiplication gives

$$Q\mathbf{1} = 0, \quad Q|_{\mathbf{1}^\perp} = -\tfrac{1}{3}I, \quad \text{rank } Q = 3. \quad (\text{FC3})$$

Thus a specific noninvertible map is already present: simultaneous altitude readout. Its entire kernel is the constant vector. This is the source's tetrahedral projection, not a singularity of WCF6.

The affine map which recovers the four indicators is

$$b : V_{\mathbb{C}} \longrightarrow \{c \in \mathbb{C}^4 : \mathbf{1}^T c = 1\}, \quad b(v) = \frac{1}{4}\mathbf{1} - \frac{3\sqrt{3}}{2}v, \quad b(\mathbf{q}_a) = e_a. \quad (\text{FC4})$$

Its inverse is $c \mapsto -2(c - \mathbf{1}/4)/(3\sqrt{3})$. Both formulas follow by substitution in FC1. They commute with all simultaneous permutations of the four labels. Retaining the affine term in FC4 recovers the constant direction that FC3 loses.

2 Exact transport into the original quartet

Retain the WCF simple quartet and its full unit, with $0 < \delta < 1/2$ and $\gamma > 2$. Fix the explicit marking

$$\begin{array}{c|cccc} a & \text{M} & \text{N} & \text{T} & \text{E} \\ \hline \rho_a & \frac{1}{2} + \delta + i\gamma & \frac{1}{2} + \delta - i\gamma & \frac{1}{2} - \delta + i\gamma & \frac{1}{2} - \delta - i\gamma. \end{array} \quad (\text{FC5})$$

This is a declared marking, not an inference that the four Time operators are the arithmetic dilation operator. All other markings are obtained by simultaneous permutations; their effects on the Segre relation are calculated below. Put $h(S) = \prod_a (S - \rho_a)$ and

$$\ell_a(S) = \prod_{b \neq a} \frac{S - \rho_b}{\rho_a - \rho_b}. \quad (\text{FC6})$$

The exact algebra map is

$$\Theta : \text{Map}(\{\mathbf{q}_a\}, \mathbb{C}) \longrightarrow \mathbb{C}[S]/(h), \quad f \mapsto \sum_a f(\mathbf{q}_a)[\ell_a]. \quad (\text{FC7})$$

Evaluation at the four ρ_a is its inverse. Indeed $\ell_a(\rho_b) = \delta_{ab}$; a polynomial of degree at most three vanishing at all four distinct points is zero. The same argument applied to remainders modulo h proves $[\ell_a \ell_b] = \delta_{ab}[\ell_a]$ and $\sum_a [\ell_a] = 1$, so FC7 is an algebra isomorphism, not merely a vector-space bijection.

For the full original Taylor unit $u_a = (g/h)(\rho_a) \neq 0$, the additional source-unit map is $c \mapsto \text{diag}(u_a)c$, with inverse $c \mapsto \text{diag}(u_a^{-1})c$. Its determinant is $\prod_a u_a$. No unit has been replaced by one in the preceding isomorphism or in the period map below. The unit multiplication is a separate linear isomorphism, not an unital algebra homomorphism.

The original exponential observation of a coefficient vector is

$$\mathcal{E}(c)(z) = \sum_a c_a e^{\rho_a z}, \quad (\mathcal{E}(c)^{(n)}(0))_{n=0}^3 = V_\rho c, \quad (V_\rho)_{na} = \rho_a^n. \quad (\text{FC8})$$

The Vandermonde determinant, with precisely the order FC5, is

$$\det V_\rho = \prod_{a < b} (\rho_b - \rho_a) = -64\delta^2\gamma^2(\delta^2 + \gamma^2) \neq 0. \quad (\text{FC9})$$

The product formula follows by subtracting columns and induction in the number of columns; its evaluation uses the six differences $-2i\gamma, -2\delta, -2\delta - 2i\gamma, -2\delta + 2i\gamma, -2\delta, -2i\gamma$. In particular the four transported points cannot collide under the full first-four-jet observation. With the unit included the determinant is FC9 times $\prod_a u_a$.

3 The degree-eight map is the original conductor map

Write $t = (t_1, t_2, t_3, t_4)$ in the FC5 order, and retain

$$Q_0(t) = t_1 t_4 - t_2 t_3, \quad t(z) = (e^{\rho_1 z}, e^{\rho_2 z}, e^{\rho_3 z}, e^{\rho_4 z}). \quad (\text{FC10})$$

Let $\mathcal{S}_n = \mathbb{C}[t_1, t_2, t_3, t_4]_n$. Substitution along this curve gives the linear map

$$\mathcal{E}_n : \mathcal{S}_n \rightarrow \text{span}\{e^{\beta_{n,u,w} z} : 0 \leq u, w \leq n\}, \quad P \mapsto P(t(z)), \quad \beta_{n,u,w} = \frac{n}{2} + (2u - n)\delta + i(2w - n)\gamma. \quad (\text{FC11})$$

Its exact kernel is

$$\ker \mathcal{E}_n = Q_0 \mathcal{S}_{n-2}, \quad \dim \text{im } \mathcal{E}_n = (n+1)^2, \quad (\text{FC12})$$

where $\mathcal{S}_{n-2} = 0$ for $n < 2$. To prove this, a monomial $t_1^{k_1} \cdots t_4^{k_4}$ is sent to the exponential indexed by $u = k_1 + k_2, w = k_1 + k_3$. Every pair $0 \leq u, w \leq n$ occurs: choose $k_1 = \max(0, u + w - n)$, $k_2 = u - k_1$, $k_3 = w - k_1$, $k_4 = n - u - w + k_1$. Replacing $t_1 t_4$ by $t_2 t_3$ reduces any monomial to this unique representative. Thus the quotient by Q_0 has those $(n+1)^2$ monomial representatives. Their exponents are distinct because equality of real parts forces $u = u'$ and equality of imaginary parts forces $w = w'$. Differentiating a linear relation in these exponentials through order $(n+1)^2 - 1$ gives a Vandermonde matrix with nonzero determinant. Hence they are linearly independent and FC12 follows. At $n = 8$, the map has rank 81 and kernel dimension $\binom{11}{3} - 81 = 84$.

For the original admitted period let H be its entire invertible period matrix, including its Gamma factors, branch powers and Taylor unit. Retain $D_5 = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and the precise original polynomials

$$Q_j(X) = Q_0(H^{-1} D_5^{-j} X), \quad A_H(X) = \prod_{j=1}^4 Q_j(X), \quad E_A(z) = A_H(Ht(z)). \quad (\text{FC13})$$

These are NC5 and DFX4–5 [4, 5]; no alternative conductor is being selected. Expanding $A_H(H(ZW, Z, W, 1)^T) = \sum_{u,w=0}^8 a_{uw} Z^u W^w$ gives exactly

$$E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_j = \sum_{u,w=0}^8 a_{uw} \beta_{8,u,w}^j. \quad (\text{FC14})$$

This follows because $t(z) = e^{(1/2 - \delta - i\gamma)z} (e^{2\delta z} e^{2i\gamma z}, e^{2\delta z}, e^{2i\gamma z}, 1)^T$ and A_H is homogeneous of degree eight.

There is also a literal point map into the original quadric:

$$\mathfrak{q}_a \xrightarrow{b} e_a \xrightarrow{H} H e_a, \quad Q_0(e_a) = Q_0(H^{-1} H e_a) = 0. \quad (\text{FC15})$$

The canonical 488-page reader [1] also supplies an explicit four-point polynomial collision in four coordinates, theorem **explicit-four-time-Jacobian-collision**. Its ordered source columns form the matrix

$$K = \begin{pmatrix} 0 & i & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1 & -1 - \sqrt{2}i & 1 - \sqrt{2}i \\ i/2 & -3i & 2\sqrt{2} + 6i & -2\sqrt{2} + 6i \\ 0 & 13 & -34 - 19\sqrt{2}i & 34 - 19\sqrt{2}i \end{pmatrix}, \quad \det K = 77\sqrt{2}i/2. \quad (\text{FC15a})$$

Expansion along the first column reduces this determinant to $i/2$ times a three-by-three determinant. Subtracting its second column from its third gives the column $(0, 2, 68)^T$; the remaining expansion is $30\sqrt{2} + 94/\sqrt{2} = 77\sqrt{2}$, proving the value. Hence the explicit linear map $x \mapsto HK^{-1}x$, with $K^{-1} = \text{adj}(K)/(77\sqrt{2}i/2)$, sends those four original collision coordinates to exactly the same four $H e_a$ in FC15. The labelled map from these four source points to

the Time locales is $Ke_a \mapsto \mathbf{q}_a$. This supplies a second, fully explicit source for FC15 without asserting that two different ambient spaces or their dynamics are identical. The companion independent derivation records all four polynomial evaluations and their exact certificate. Put $M_j = H^{-1}D_5^{-j}H$. The exact conductor value at this point is

$$A_H(He_a) = \prod_{j=1}^4 ((M_j)_{1a}(M_j)_{4a} - (M_j)_{2a}(M_j)_{3a}). \quad (\text{FC16})$$

Its vanishing means that He_a lies on another quadric in the orbit. It is not the same evaluation as any μ_v : the latter is the derivative at $z = 0$ on the path $Ht(z)$, whose base point is $H\mathbf{1}$. Equations FC13–16 give both maps and their different evaluation points explicitly, rather than discarding the relation between the objects.

More precisely, the four incidences are four coordinates of the *original* symbol, not an auxiliary symbol. With $P(t) = A_H(Ht) = \sum_{|m|=8} c_m t^m$, collection in FC11 gives

$$a_{u,w} = \sum_{\substack{|m|=8 \\ m_1+m_2=u, m_1+m_3=w}} c_m, \quad (A_H(He_1), A_H(He_2), A_H(He_3), A_H(He_4)) = (a_{88}, a_{80}, a_{08}, a_{00}). \quad (\text{FC16a})$$

Indeed each of those four corner pairs admits only the corresponding pure eighth power. Define $\mathcal{D}_\rho = \sum_j \rho_j t_j \partial_{t_j}$. The chain rule, iterated on each polynomial $\mathcal{D}_\rho^n P$, proves

$$\mu_n = (\mathcal{D}_\rho^n P)(\mathbf{1}) = \sum_{j=1}^4 A_H(He_j) (8\rho_j)^n + \sum_{(u,w) \notin \{(8,8), (8,0), (0,8), (0,0)\}} a_{uw} \beta_{8,u,w}^n. \quad (\text{FC16b})$$

Thus all four points enter the exact moment test, with the remaining 77 collected terms explicitly retained. Computing those four incidences alone does not evaluate the remaining sum. For example, the test polynomial $t_1^7 t_2$ has zero at all four e_j and value one at $\mathbf{1}$; this establishes independence of the evaluation functionals, without substituting that polynomial for the original conductor.

On the original five-orbit domain the quadrics $Q_0(H^{-1}X), Q_1, \dots, Q_4$ are nonassociate irreducibles. The polynomial $Q_0(t)$ is irreducible: as a degree-one polynomial in t_1 it is primitive over $\mathbb{C}[t_2, t_3, t_4]$, with coprime coefficients t_4 and $-t_2 t_3$, and is irreducible over that coefficient field. An invertible linear substitution preserves irreducibility. If the degree-eight polynomial $A_H(Ht)$ belonged to (Q_0) , primality would force Q_0 to divide one factor $Q_j(Ht)$, making those degree-two polynomials associates. This contradicts the original orbit domain. Thus FC12 proves

$$E_A \not\equiv 0, \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0. \quad (\text{FC17})$$

The order bound follows again from the 81 distinct exponents and their first 81 derivative matrix. It is not a numerical lower bound for the first nonzero derivative. More precisely, if r_A is the number of nonzero original collected coefficients, the same derivative argument gives $v \leq r_A - 1$. In particular, vanishing of all four incidences in FC16a leaves $r_A \leq 77$ and gives $v \leq 76$; it does not annihilate the original nonzero symbol. This support statement evaluates the effect of all four point-incidence equations on the vanishing-order bound.

4 All tetrahedral permutations, with the lost equation restored

The full S_4 action on the four labels takes the line $\mathbb{C}Q_0$ to the three lines represented by

$$Q_{14|23} = t_1 t_4 - t_2 t_3, \quad Q_{12|34} = t_1 t_2 - t_3 t_4, \quad Q_{13|24} = t_1 t_3 - t_2 t_4. \quad (\text{FC18})$$

These correspond to the three partitions into two unordered pairs. The stabilizer of the first line has order eight (independent exchanges inside the pairs and exchange of the pairs); it is the dihedral group of the rectangle. A sign can change when the pairs are exchanged and is retained by these equations. The three lines are distinct, since their monomial supports differ. On the unchanged original curve,

$$Q_{14|23}(t(z)) = 0, \quad Q_{12|34}(t(z)) = 2e^z \sinh(2\delta z), \quad Q_{13|24}(t(z)) = 2ie^z \sin(2\gamma z). \quad (\text{FC19})$$

All identities follow by multiplying the original exponentials. The last two have nonzero first derivatives 4δ and $4i\gamma$ at zero. Thus equal-looking tetrahedral presentations must carry their pairing equation as well as their point labels. To transport the whole diagram under a permutation matrix P , use $t' = Pt$, $H' = HP^{-1}$ and $Q'_0(t') = Q_0(P^{-1}t')$. Then $H't' = Ht$ and $Q'_0(H'^{-1}X) = Q_0(H^{-1}X)$ exactly; the original conductor is unchanged. Permuting only labels while silently fixing a different pairing would instead change its defining equation.

5 The singularity test in equation (11)

Retain $d = D + 1$, $s \in \{1, a\}$ with $a = 4\ell + 1 \geq 9$, $b = s/2$, and $N = D + v$. WCF2–6 specify the original Gamma polynomial norms and their orthonormal frames; write

$$\rho_n^{(s)} = \sqrt{n!/(b)_n}, \quad w(t) = -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g_0 = (-i)^v \mu_v / v!. \quad (\text{FC20})$$

The source and target of the unscaled map are exactly

$$T_A : \mathcal{P}_{D+v}/\mathcal{P}_{v-1} \rightarrow \mathcal{P}_D, \quad T_A P(S') = \sum_{u,w} a_{uw} P(S' + \beta_{8,u,w}), \quad (\text{FC21})$$

with centers $a/2$ and $a/2 - 4$ and the WCF2 quotient/target norms. Its WCF6 matrix and determinant are

$$A_{D,s} = \text{diag}(\rho_0^{(s)}, \dots, \rho_D^{(s)})^{-1} T_D(g) \text{diag}(\rho_v^{(s)}, \dots, \rho_{D+v}^{(s)}), \quad (\text{FC22})$$

$$\det A_{D,s} = \left(\frac{(-i)^v \mu_v}{v!} \right)^{D+1} \prod_{j=0}^D \frac{\rho_{j+v}^{(s)}}{\rho_j^{(s)}} \neq 0. \quad (\text{FC23})$$

Indeed Taylor expansion gives $T_A S^n = \sum_{j=v}^n \binom{n}{j} \mu_j S'^{n-j}$. It annihilates precisely $\mathcal{P}_{v-1}$ and reduces each remaining leading degree by v with nonzero factor $\binom{n}{v} \mu_v$. Successive elimination proves bijectivity in FC21. In the stated frames FC22 is triangular; its diagonal proves FC23. Since $b > 0$, all displayed weights are finite and positive. Nothing about the four labelled points can defeat FC23 while retaining the stated orbit domain and actual vanishing order.

For completeness, every inverse coefficient is explicitly reconstructed:

$$h_0 = g_0^{-1}, \quad h_n = -g_0^{-1} \sum_{j=1}^n g_j h_{n-j}, \quad A_{D,s}^{-1} = \text{diag}(\rho_v^{(s)}, \dots, \rho_{D+v}^{(s)})^{-1} \cdot T_D(h) \text{diag}(\rho_0^{(s)}, \dots, \rho_D^{(s)}). \quad (\text{FC24})$$

Multiplication of the two truncated Toeplitz matrices gives the identity by the convolution recurrence, proving this inverse with all factors.

There is a precise rank-drop locus for a *fixed* old quotient. Keep v_0 and $N = D + v_0$ fixed while varying coefficients and shifts within the conductor formula. It defines $\mathcal{P}_N/\mathcal{P}_{v_0-1} \rightarrow \mathcal{P}_D$ exactly when $\mu_0 = \dots = \mu_{v_0-1} = 0$. Necessity follows by applying the map to $1, S, \dots, S^{v_0-1}$ in that order; sufficiency follows from its Taylor formula. At such a point of finite actual order $v \geq v_0$,

$$\text{rank } T_A = \max(D + 1 - (v - v_0), 0), \quad \ker T_A = \mathcal{P}_{\min(v-1, N)}/\mathcal{P}_{v_0-1}. \quad (\text{FC25})$$

To prove this, degrees below v vanish and the images of $S^v, \dots, S^N$ have different degrees and nonzero leading coefficients; they span $\mathcal{P}_{N-v}$ when $N \geq v$. If $E_A = 0$ identically the rank is zero. If actual order is smaller than v_0 , the old quotient arrow is undefined, not singular. Restoring the actual kernel and actual target degree gives FC21 again. Thus a zero of the old leading moment is an order jump, and the full kernel correction must be retained.

The audit's inverse-exterior expression (11) has singular factors only when the retained leading moment vanishes or a displayed weight ceases to be defined. The latter never occurs for the admitted positive s ; the former contradicts using the actual order v . FC25 states exactly what a genuine fixed-quotient degeneration would be. No original-period order-jump point is supplied merely by the four Fable locale coordinates. The point-conductor incidence test FC16 remains an explicit separate polynomial calculation in the original H , with all four columns retained.

6 Four explicit inputs to the very operator in equation (11)

For the programme cutoffs $D \geq 3$, the four labels have the following literal receiving vectors in the WCF source, without introducing another conductor:

$$x_a = [S^v \ell_a(S)] \in \mathcal{P}_{D+v}/\mathcal{P}_{v-1}, \quad y_a(S') = \sum_{u,w=0}^8 a_{uw}(S' + \beta_{8,u,w})^v \ell_a(S' + \beta_{8,u,w}). \quad (\text{FC26})$$

Thus $\mathbf{q}_a \mapsto x_a \mapsto y_a = T_A x_a$ is an explicit map into equation (11)'s actual operator; the four-dimensional collision coordinates Ke_a map to the same x_a by first applying K^{-1} . All four output polynomials lie in $\mathcal{P}_3$. With the source label basis in FC5 and target monomial basis $(1, S', S'^2, S'^3)$, their four-column determinant is

$$\det([y_1] [y_2] [y_3] [y_4]) = \frac{\mu_v^4 \prod_{j=0}^3 \binom{v+j}{v}}{-64\delta^2\gamma^2(\delta^2 + \gamma^2)} \neq 0. \quad (\text{FC27})$$

Here the brackets on the left denote coefficient columns, not an orthonormal change of the WCF metric. To prove the formula, evaluation of a cubic at the four roots has matrix V_ρ^T . Therefore the coefficient matrix of the four ℓ_a is $(V_\rho^T)^{-1}$. On the source monomials $S^v, \dots, S^{v+3}$, Taylor expansion in FC21 gives a triangular output matrix with diagonal $\mu_v \binom{v+j}{v}$. Multiplying these two determinants proves FC27. It also proves all four outputs are linearly independent. The source norm is the original quotient norm of the classes in FC26; choosing their minimum-norm representatives only subtracts elements of $\mathcal{P}_{v-1}$, which T_A kills exactly. The target norm is the unchanged original Gamma norm at center $a/2 - 4$. Thus no modification of the norms is used to establish the determinant or the images.

The full four-vector Gram determinants can also be evaluated, so the norm transport need not be left implicit. Set $\Delta_\rho = -64\delta^2\gamma^2(\delta^2 + \gamma^2)$ and retain

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = (2\pi)^{s/2} n!(s/2)_n. \quad (\text{FC28})$$

Let $G_x = (\langle x_a, x_b \rangle_{s,a/2,\text{quot}})_{ab}$ and $G_y = (\langle y_a, y_b \rangle_{s,a/2-4})_{ab}$. Then

$$\det G_x = \frac{\prod_{j=0}^3 h_{v+j,s}}{|\Delta_\rho|^2}, \quad \det G_y = \frac{|\mu_v|^8 \left(\prod_{j=0}^3 \binom{v+j}{v} \right)^2 \prod_{j=0}^3 h_{j,s}}{|\Delta_\rho|^2}, \quad (\text{FC29})$$

and in particular

$$\frac{\det G_y}{\det G_x} = \left| \frac{\mu_v}{v!} \right|^8 \prod_{j=0}^3 \frac{(\rho_{v+j}^{(s)})^2}{(\rho_j^{(s)})^2} = |\det A_{3,s}|^2 > 0. \quad (\text{FC30})$$

For the proof, the original orthonormal polynomial of degree n in S has leading coefficient $i^{-n}/\sqrt{h_{n,s}}$, as in WCF2. Expanding $S^v, \dots, S^{v+3}$ in this frame and removing the degrees below v gives a triangular matrix with diagonal $i^{v+j}\sqrt{h_{v+j,s}}$. Its Gram determinant is $\prod_{j=0}^3 h_{v+j,s}$. Multiplication by $(V_\rho^T)^{-1}$ therefore proves the first formula in FC29. In the target, the monomial Gram determinant is $\prod_{j=0}^3 h_{j,s}$ by the same leading-coefficient argument; FC27 proves the second formula. Dividing the two equalities, substituting $h_{n,s}$ and the binomial coefficients, proves FC30. Both Gram determinants retain the mass $(2\pi)^{2s}$; it cancels only in their displayed ratio. This is the original quotient metric on the four vectors, not a Euclidean norm imposed on their coefficient columns.

These formulas locate the noninvertibility that survives transport: FC3 loses the constant direction when reading the four locales together, and the collision polynomial forgets which marked source was chosen. FC26 instead retains that source identity through the original conductor; FC27 excludes a four-point collision or a kernel there. An attempt to factor FC26's four outputs through a single common image would make all four columns equal, contradicting FC27. This proves the failure of that specific factorization with the exact source and target maps in hand, rather than asserting that the programmes are unrelated.

7 Transport of the complete nonlinear collision

In this section and the next two sections, a is the first ES coordinate. It is separate from the fixed WCF amplification integer, which is now written $a_{\text{amp}} = 4\ell + 1 \geq 9$. Thus the original source and target centers remain $a_{\text{amp}}/2$ and $a_{\text{amp}}/2 - 4$, and $s \in \{1, a_{\text{amp}}\}$. This dictionary changes no coordinate value or measure. Here the receiving map is constructed, rather than inferred from a similarity of determinants. In the original source coordinates $q = (a, y, z, w)$, the ES reader's four-dimensional polynomial is

$$\begin{aligned} P_0(q) &= a^3 z + 2a^2 y - ia, \\ P_2(q) &= -a^3 y^2 z - 2ia^2 yz + a^2 w - 2a^2 y^3 - 10ia y^2 + 3az + y, \\ P_3(q) &= 2a^3 y^3 z + 6ia^2 y^2 z + 2a^2 wy + 4a^2 y^4 + 2iaw - 4ia y^3 - 2ayz + 2iz + 7y^2, \\ P_4(q) &= 2a^3 y^4 z + 8ia^2 y^3 z + a^2 wy^2 + 4a^2 y^5 + 2iawy + 7ia y^4 - 10ay^2 z - 4iyz - w - 3y^3. \end{aligned} \tag{FC31}$$

Write $P = (P_0, P_2, P_3, P_4)$ and $h = (0, 0, -1, 0)^T$. Substitution in FC31 gives $P(Ke_a) = h$ for the four columns of FC15a, and $h = 2iKe_M$. Direct differentiation and determinant expansion give $\det DP = -2$ identically; the accompanying `independent/verify_four_labels.py` performs that exact polynomial calculation and all four evaluations, with rational coefficients in $\mathbb{Q}(i, \sqrt{2})$.

Let $\mathcal{U} = \mathcal{P}_{v+3}/\mathcal{P}_{v-1}$ and $\mathcal{W} = \mathcal{P}_3$, as subspaces of FC21, and use the original quotient and target Gamma norms. Define the isomorphisms

$$X : \mathbb{C}^4 \rightarrow \mathcal{U}, \quad Xe_a = x_a, \quad Y : \mathbb{C}^4 \rightarrow \mathcal{W}, \quad Ye_a = y_a, \quad \Phi = XK^{-1}, \quad \Psi = YK^{-1}. \tag{FC32}$$

FC27 proves they are isomorphisms and $T_A \Phi = \Psi$. Consequently the following are explicit polynomial self-maps of the *actual* original four-dimensional source and target spaces:

$$P_{\mathcal{U}} = \Phi P \Phi^{-1}, \quad P_{\mathcal{W}} = \Psi P \Psi^{-1}, \quad T_A P_{\mathcal{U}} = P_{\mathcal{W}} T_A. \tag{FC33}$$

The last equality follows by substituting $T_A \Phi = \Psi$ and $\Phi^{-1} = \Psi^{-1} T_A$ on $\mathcal{U}$. Both polynomial maps have constant Jacobian determinant -2 : the chain rule cancels the two constant change-of-coordinate determinants. Their norms remain those of FC28–30. The pulled-back forms on $\mathbb{C}^4$ are, explicitly, $(K^{-1})^* G_x K^{-1}$ and $(K^{-1})^* G_y K^{-1}$.

In particular the four distinct original target polynomials satisfy

$$P_{\mathcal{W}}(y_M) = P_{\mathcal{W}}(y_N) = P_{\mathcal{W}}(y_T) = P_{\mathcal{W}}(y_E) = 2i y_M. \tag{FC34}$$

This is a transported noninjective nonlinear map on the conductor target. The original linear conductor is the invertible intertwiner in FC33; it has not acquired a kernel and none of its coefficients has been fitted to the new map.

No inverse is hidden in these definitions: put $\ell_a(S) = \sum_{r=0}^3 \lambda_{ra} S^r$, where its coefficients are given by the product FC6. In the target monomial basis the matrix of Y is exactly

$$Y_{ka} = \sum_{r=k}^3 \lambda_{ra} \binom{v+r}{k} \mu_{v+r-k}, \quad 0 \leq k \leq 3, \quad P_{\mathcal{W}}(p) = YK^{-1}P(KY^{-1}[p]). \quad (\text{FC35})$$

Here $[p]$ is the coefficient column of p ; the last output column is read in the same monomial basis. The inverses are adjugates divided by their nonzero determinants in FC15a and FC27. Formula FC35, together with FC31, is the complete polynomial map in the original target coordinates, with all original moments retained.

8 The full fibre contains seven points, with all four labels retained

The four displayed source points of FC15a are not the entire polynomial fibre. This does not identify or remove any of the four Time locales. We now solve every equation $P(q) = h$ over $\mathbb{C}$.

When $a = 0$, FC31 reads

$$P(q) = (0, y, 2iz + 7y^2, -4iyz - w - 3y^3).$$

Thus $y = 0, z = i/2, w = 0$, the single point $q_M = Ke_M$. When $a \neq 0$, the equations $P_0 = P_2 = 0$ solve uniquely as

$$z = \frac{i - 2ay}{a^2}, \quad w = \frac{i(7a^2y^2 - 3iay - 3)}{a^3}. \quad (\text{FC36})$$

Substitution into $P_3 + 1$ and P_4 gives exactly

$$P_3 + 1 = -\frac{3a^2y^2 - a^2 + 6iay - 4}{a^2}, \quad P_4 = -\frac{(ay + i)(2a^2y^2 + 4iay - 3)}{a^3}. \quad (\text{FC37})$$

Set $t = ay$. Hence

$$(t + i)(2t^2 + 4it - 3) = 0, \quad a^2 = 3t^2 + 6it - 4. \quad (\text{FC38})$$

There are exactly three possible t : $-i$ and $-i \pm 1/\sqrt{2}$. For $t = -i$ the second equation gives $a^2 = -1$; for the other two it gives $a^2 = 1/2$. In each case both square roots are nonzero, $y = t/a$ is determined, and FC36 determines z, w . Thus there are exactly six points with $a \neq 0$ and the one already found with $a = 0$. All implications used above are equivalences, so no elimination root is extraneous and none is omitted.

The exact involution

$$J(a, y, z, w) = (-a, -y, z, -w)$$

satisfies, term by term in FC31,

$$P \circ J = J_{\text{out}} \circ P, \quad J_{\text{out}}(u_0, u_2, u_3, u_4) = (-u_0, -u_2, u_3, -u_4). \quad (\text{FC39})$$

Since $J_{\text{out}}h = h$, the full fibre is

$$P^{-1}(h) = \{q_M, q_N, q_T, q_E, Jq_N, Jq_T, Jq_E\}. \quad (\text{FC40})$$

The three additional points, in the unchanged (a, y, z, w) coordinates, are

$$\begin{aligned} Jq_N &= (-i, 1, -3i, -13), \\ Jq_T &= (-1/\sqrt{2}, 1 + \sqrt{2}i, 2\sqrt{2} + 6i, 34 + 19\sqrt{2}i), \\ Jq_E &= (-1/\sqrt{2}, -1 + \sqrt{2}i, -2\sqrt{2} + 6i, -34 + 19\sqrt{2}i). \end{aligned} \quad (\text{FC41})$$

Their first and second coordinates distinguish all seven, and FC36–38 prove their exhaustiveness.

This entire fibre transports to the original WCF target. Since $Jq = -q + 2ze_z$ and $e_z = -2iq_M$, its seven points are

$$\begin{aligned} & y_M, y_N, y_T, y_E, \quad -12y_M - y_N, \\ & (24 - 8\sqrt{2}i)y_M - y_T, \quad (24 + 8\sqrt{2}i)y_M - y_E. \end{aligned} \quad (\text{FC42})$$

They are precisely $P_{\mathcal{W}}^{-1}(2iy_M)$, by the bijective conjugacy FC33. Every one is a locally invertible point of $P_{\mathcal{W}}$ because its Jacobian determinant is -2 . The failure is a global inverse at the common target, not a zero of the local derivative or of the linear conductor determinant.

9 The explicit escaping branch and its original norm

There is also a rational symmetry which leaves every output unchanged. On the exact domain $a \neq 0$, set

$$\Sigma(a, y, z, w) = \left(-a, y + \frac{2i}{a}, \frac{6i}{a^2} - z, w - \frac{14iy^2}{a} + \frac{28y}{a^2} + \frac{40i}{a^3} \right). \quad (\text{FC43})$$

Direct substitution gives $\Sigma^2 = I$ and $P\Sigma = P$ on this domain. One can verify the latter identity with fewer terms in the reader's original factor coordinates:

$$\begin{aligned} b &= i + ay, & c &= -i + 2ay + a^2z, \\ d &= -iy - a(iz + 2y^2) - a^2yz, & e &= 2z - 7iy^2 + aw, \\ f &= iw + 3iy^3 - 4yz + a(6iy^2z + wy + 4y^4) + 2a^2y^3z. \end{aligned} \quad (\text{FC44})$$

Substitution of FC43 into these formulas changes (a, b, c, d, e, f) to $(-a, -b, -c, -d, -e, -f)$. The polynomial outputs are $(ac, ae + bd, af + be, bf)$, so all four remain unchanged. The first four solved coordinates also show the sign operation is an involution; alternatively it follows by inserting FC43 twice. For $a = 0$, the chart fixes $b = i$. The negative of the M factor state has $b = -i$ and lies outside this chart. The other three factor-state negatives are present among FC41. This records the orientation data without treating an opposite factorization as an identical labelled point.

The exact factor count explains why four labels coexist with seven points in this affine fibre. A marked factorization of $H(U, V) = UV(U - V)(U + V)$ chooses one of its four distinct linear factors as $L = aU + bV$, and places the remaining factors in $C = cU^3 + dU^2V + eUV^2 + fV^3$. The resultant is

$$R(L, C) = C(-b, a) = a^3f - a^2be + ab^2d - b^3c. \quad (\text{FC44a})$$

For each marked root it is nonzero, because none of the remaining three factors vanishes on that root. Replacing (L, C) by $(\lambda L, \lambda^{-1}C)$ changes this resultant by λ^2 . Thus $R = 1$ has exactly two scalar choices for each of the four marked roots, eight factor states in all. Their signs are distinct. All have $ad + bc = 1$, the U^3V coefficient of H .

On $a \neq 0$, FC44 has the exact inverse

$$y = (b - i)/a, \quad z = (c + i - 2ay)/a^2, \quad w = (e - 2z + 7iy^2)/a. \quad (\text{FC44b})$$

The incidence equation $ad + bc = 1$ then determines d uniquely, and FC44a with $R = 1$ determines f uniquely. This proves that all six factor states with $a \neq 0$ occur in the chart. When $a = 0$, those two incidence equations force $bc = 1$ and $-b^3c = 1$, hence $b^2 = -1$. The chart FC44 has $b = i$ at $a = 0$; its remaining coordinates are recovered by $y = id$, $z = (e + 7iy^2)/2$ and $w = (f - 3iy^3 + 4yz)/i$. It contains the entire $b = i$ component and none of the $b = -i$ component. At the particular product H , the omitted state is $(a, b, c, d, e, f) = (0, -i, i, 0, -i, 0)$. This proves

precisely that the chosen chart retains seven of the eight signed factor states, still over the same four marked roots. No Time label has been added or discarded.

Take real $\epsilon > 0$ and the two explicit source curves

$$q_\epsilon = (\epsilon, 0, i/2, 0), \quad \hat{q}_\epsilon = \Sigma q_\epsilon = (-\epsilon, 2i/\epsilon, 6i/\epsilon^2 - i/2, 40i/\epsilon^3). \quad (\text{FC45})$$

The latter escapes to infinity whereas the former tends to q_M . Their identical images are

$$P(q_\epsilon) = P(\hat{q}_\epsilon) = (-i\epsilon + i\epsilon^3/2, 3i\epsilon/2, -1, 0) \longrightarrow h. \quad (\text{FC46})$$

Thus h is an explicit nonproper value of the nonlinear map: a sequence escaping every compact subset has convergent images. No numerical search is needed.

The same statement holds in the original target norm, with exact constants. Put $z_\epsilon = \Psi \hat{q}_\epsilon$ and

$$R = \Psi e_w = \frac{6}{77}y_M + \frac{1}{77}y_N + \frac{-3 - \sqrt{2}i}{154}y_T + \frac{3 - \sqrt{2}i}{154}y_E. \quad (\text{FC47})$$

The displayed coefficients are the last column of K^{-1} , verified by multiplying by FC15a. They are not all zero, so $R \neq 0$ by FC27. Multiplying FC45 by $\epsilon^3 \Psi$ gives the norm-convergent limit

$$\epsilon^3 z_\epsilon \longrightarrow 40iR, \quad \epsilon^3 \|z_\epsilon\|_{s, a_{\text{amp}}/2-4} \longrightarrow 40\|R\|_{s, a_{\text{amp}}/2-4} > 0. \quad (\text{FC48})$$

The limit uses only a fixed finite-dimensional linear map and the original positive Gamma norm. Simultaneously

$$P_W(z_\epsilon) = \Psi(-i\epsilon + i\epsilon^3/2, 3i\epsilon/2, -1, 0) \longrightarrow 2iy_M. \quad (\text{FC49})$$

Equations FC42 and FC45–49 therefore identify the full finite fibre and a branch escaping to infinity above its common value in the actual conductor target. They do not make that introduced nonlinear map a linear conductor or an arithmetic dilation action. The exact relation to the original conductor is the intertwining square FC33, which preserves its inverse bound.

10 Compatibility with the original translation action

The nonlinear transport can be tested against an actual symmetry of the original conductor. It is not enough to have a conjugate polynomial on its target. The original finite-shift formula FC21 satisfies

$$T_A \partial_S = \partial_{S'} T_A, \quad T_A(P(S+t)) = (T_A P)(S' + t). \quad (\text{FC50})$$

Both identities follow by differentiating, respectively translating, each term of the unchanged finite sum. Differentiation preserves $\mathcal{P}_{v-1}$ and therefore descends on $\mathcal{U}$. In the source basis $[S^v], \dots, [S^{v+3}]$ and target basis $1, S', S'^2, S'^3$, its matrices are

$$N_v = \begin{pmatrix} 0 & v+1 & 0 & 0 \\ 0 & 0 & v+2 & 0 \\ 0 & 0 & 0 & v+3 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad N_0 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{FC51})$$

Let B_v have entries $(B_v)_{kj} = \binom{v+j}{k} \mu_{v+j-k}$ for $k \leq j$ and zero otherwise. FC35 gives $Y = B_v(V_\rho^T)^{-1}$, and FC50 gives $N_0 B_v = B_v N_v$. Thus the complete original translation matrix in the Fable coordinates is exactly

$$M = \Psi^{-1} N_0 \Psi = K V_\rho^T N_v (V_\rho^T)^{-1} K^{-1}, \quad M^4 = 0, \quad M^3 \neq 0. \quad (\text{FC52})$$

No moment has been set to one in this formula: their cancellation is the proved identity $N_0 B_v = B_v N_v$. They remain in Ψ and in the native metrics.

The quotient correction has an explicit form in the labelled basis. Writing $R_v = V_\rho^T N_v (V_\rho^T)^{-1}$ gives

$$(R_v)_{ba} = \ell'_a(\rho_b) + v \frac{\delta_{ba} - \ell_a(0)}{\rho_b}. \quad (\text{FC53})$$

Indeed, modulo $\mathcal{P}_{v-1}$,

$$\partial_S(S^v f(S)) = S^v \left(f'(S) + v \frac{f(S) - f(0)}{S} \right).$$

The removed term is exactly $vf(0)S^{v-1}$, and the quotient in parentheses is a polynomial. Evaluating at the four nonzero ρ_b proves FC53. It retains the term that would be lost by treating the quotient as an ordinary cubic space before transporting its action.

The introduced nonlinear map does not preserve this original translation flow. From FC31,

$$P(0) = 0, \quad J_0 := DP(0) = \text{diag}(-i, 1, 2i, -1). \quad (\text{FC54})$$

All four eigenvalues of J_0 are distinct. A matrix commuting with this diagonal matrix must be diagonal, since its (j, k) entry is multiplied in the commutator by the nonzero difference of the two eigenvalues whenever $j \neq k$. If M commuted with J_0 , it would consequently be diagonal; $M^4 = 0$ would then force $M = 0$, contradicting $M^3 \neq 0$. Hence $J_0 M - M J_0 \neq 0$.

More strongly, $P e^{tM} = e^{tM} P$ fails as a polynomial identity for every $t \neq 0$ in $\mathbb{C}$. Otherwise differentiation at zero would make J_0 commute with $R = e^{tM}$. Since $(R - I)^4 = 0$,

$$M = t^{-1} \left((R - I) - \frac{1}{2}(R - I)^2 + \frac{1}{3}(R - I)^3 \right).$$

This identity follows by substituting $R = I + tM + t^2 M^2/2 + t^3 M^3/6$ and collecting powers of M . It would force J_0 to commute with M , the contradiction just proved. Conjugating by Ψ proves the same failure for $P_{\mathcal{W}}$ and the original target translations e^{tN_0} .

The entire correction is explicit, rather than an unspecified failure:

$$\Delta(q) := DP(q)Mq - MP(q), \quad P(e^{tM}q) - e^{tM}P(q) = \int_0^t e^{(t-s)M} \Delta(e^{sM}q) ds. \quad (\text{FC55})$$

For complex t , integrate on the straight segment from zero to t . Differentiating $e^{(t-s)M}P(e^{sM}q)$ proves FC55 by the fundamental theorem of calculus. Every exponential truncates after its cubic term, so the integrand and integral are finite explicit polynomials. The linear part of Δ is precisely $(J_0 M - M J_0)q \neq 0$. The original-target correction is Ψ times the right side of FC55, with $q = \Psi^{-1}p$; its norm is evaluated in the unchanged form $(K^{-1})^* G_y K^{-1}$.

Thus FC33 is a complete nonlinear intertwining with the original conductor, FC42 is its complete seven-point collision, FC48–49 is its exact nonproper branch, and FC55 is its nonzero correction for the conductor's original translation action. None of these calculations identifies this additional nonlinear map with the arithmetic dilation action or makes the inverse-exterior bound false.

11 Every exterior inverse along the escaping branch, in the original metric

The original branch FC45 yields an exact pointwise differential calculation, not only an unbounded sequence. Throughout this section $\epsilon > 0$, the quartet, original conductor, a_{amp} , and s are fixed. Let $\hat{q}_\epsilon$ be FC45, $z_\epsilon = \Psi \hat{q}_\epsilon$, and retain the full positive matrix

$$G = (K^{-1})^* G_y K^{-1}, \quad \|q\|_G^2 = q^* G q = \|\Psi q\|_{s, a_{\text{amp}}/2-4}^2. \quad (\text{FC56})$$

No entry of this form is replaced by a Euclidean coordinate norm. Its entries can be obtained by a finite formula in the original moments, so FC56 does not leave an unspecified metric to be chosen. Write $F = YK^{-1}$ in the target monomial basis, $c = a_{\text{amp}}/2 - 4$, $b = s/2$, and $M_s = (2\pi)^{s/2}$. Then

$$G = F^* \mathcal{H} F, \quad \mathcal{H}_{jk} = \sum_{p=0}^j \sum_{q=0}^k \binom{j}{p} \binom{k}{q} c^{j+k-p-q} (-i)^p i^q m_{p+q}, \quad 0 \leq j, k \leq 3, \quad (\text{FC56a})$$

where $m_0 = M_s$, $m_2 = M_s b$, $m_4 = M_s(3b^2 + 2b)$, $m_6 = M_s(15b^3 + 30b^2 + 16b)$, and the odd moments vanish. To prove this formula, expand $(c - iy)^j (c + iy)^k$ in the defining measure FC28. The measure is even. Integrating the WCF generating function in WCF3, cited in [3], leaves only its constant orthogonal-polynomial coefficient, so its moment generating function at $u = \arctan t$ is $\int e^{uy} dm_s(y) = M_s (\cos u)^{-b}$ near zero. The coefficients through order six give exactly the stated moments by differentiation. The identity can equivalently be read coefficientwise through degree six, using orthogonality, so no exchange of an infinite series outside its convergence domain is needed. FC35 specifies every entry of F in $\mu_v, \dots, \mu_{v+3}$ and the unchanged quartet. Differentiating the polynomial FC31 and substituting FC45 gives

$$J_\epsilon = DP(\hat{q}_\epsilon) = \begin{pmatrix} 9i - 3i\epsilon^2/2 & 2\epsilon^2 & -\epsilon^3 & 0 \\ -7i/2 - 30i/\epsilon^2 & \epsilon^2 - 27 & -3\epsilon & \epsilon^2 \\ -2/\epsilon + 104/\epsilon^3 & -i\epsilon - 56i/\epsilon & -2i & 2i\epsilon \\ -4i/\epsilon^2 + 64i/\epsilon^4 & 2 + 28/\epsilon^2 & 0 & -1 \end{pmatrix}, \quad \det J_\epsilon = -2. \quad (\text{FC57})$$

Thus $DP_{\mathcal{W}}(z_\epsilon) = \Psi J_\epsilon \Psi^{-1}$. The inverse matrix, including all its lower-order terms, is

$$\begin{pmatrix} \frac{3i\epsilon^2}{2} - i & -\frac{3i\epsilon^4}{2} + 2i\epsilon^2 & \frac{3\epsilon^5}{2} - \frac{5\epsilon^3}{2} & \frac{3i\epsilon^6}{2} - 3i\epsilon^4 \\ -\frac{3}{2} + \frac{2}{\epsilon^2} & \frac{3\epsilon^2}{2} - 3 & \frac{3i\epsilon^3}{2} - \frac{7i\epsilon}{2} & -\frac{3\epsilon^4}{2} + 4\epsilon^2 \\ \frac{9\epsilon}{4} - \frac{18}{\epsilon} + \frac{12}{\epsilon^3} & -\frac{9\epsilon^3}{4} + \frac{39\epsilon}{2} - \frac{24}{\epsilon} & -\frac{9i\epsilon^4}{4} + \frac{81i\epsilon^2}{4} - \frac{59i}{2} & \frac{9\epsilon^5}{4} - 21\epsilon^3 + 35\epsilon \\ 3 - \frac{138}{\epsilon^2} + \frac{120}{\epsilon^4} & -3\epsilon^2 + 140 - \frac{212}{\epsilon^2} & -3i\epsilon^3 + 141i\epsilon - \frac{258i}{\epsilon} & 3\epsilon^4 - 142\epsilon^2 + 303 \end{pmatrix}. \quad (\text{FC58})$$

This is J_ϵ^{-1} : multiplying FC57 by FC58 gives the identity entry by entry. It is also $-\text{adj}(J_\epsilon)/2$. These two exact checks, rather than numerical singular-value sampling, are recorded by `derive_escape_differential.py`.

For exterior powers use increasing coordinate subsets in the order $1 = a, 2 = y, 3 = z, 4 = w$. Let $E_{I,J}^{(r)}$ send the coordinate wedge e_J to e_I and annihilate the other coordinate wedges. Expanding the minors of FC57 proves all four finite matrix limits

$$\begin{array}{c|c|c|c} r & m_r & \lim_{\epsilon \downarrow 0} \epsilon^{m_r} \wedge^r J_\epsilon & \lim_{\epsilon \downarrow 0} \epsilon^{m_r} \wedge^r J_\epsilon^{-1} \\ \hline 1 & 4 & 64i E_{4,1}^{(1)} & 120 E_{4,1}^{(1)} \\ 2 & 5 & -672 E_{34,12}^{(2)} & 336 E_{34,12}^{(2)} \\ 3 & 4 & 240 E_{234,123}^{(3)} & 32i E_{234,123}^{(3)} \\ 4 & 0 & -2 & -1/2 \end{array}. \quad (\text{FC59})$$

For explicit verification, each exterior matrix has entries $\det(J_{I,J})$ or $\det((J^{-1})_{I,J})$. Their minimum Laurent exponents are respectively $-4, -5, -4, 0$; the coefficients at those exponents are exactly the displayed single-entry matrices. Every remaining monomial has larger exponent. This proves the limits and the absence of any more singular omitted entry. The accompanying script expands every minor and checks the limits symbolically over $\mathbb{Q}(i)(\epsilon)$.

Here are the exact constants in the original norm. For a principal submatrix write $G_{I,I}$; inverses below are of the full matrix G . Set

$$\tau_1 = \sqrt{G_{44}(G^{-1})_{11}}, \quad \tau_2 = \sqrt{\det G_{34,34} \det(G^{-1})_{12,12}} = \frac{\det G_{34,34}}{\sqrt{\det G}} > 0. \quad (\text{FC60})$$

Both constants are strictly positive because G is positive definite. For the original wedge metric, the squared norm of e_I is $\det G_{I,I}$ and that of its coordinate dual functional e_J^* is $\det(G^{-1})_{J,J}$. The norm of a rank-one map $x\lambda$ is $\|x\|\|\lambda\|$: the inequality is Cauchy–Schwarz for the Riesz representative of λ , and equality is attained at that representative. It follows that the rank-one norms in FC59 are $64\tau_1$, $672\tau_2$, and $240\tau_1$. For the last equality use

$$\det G_{234,234} = \det G (G^{-1})_{11}, \quad \det(G^{-1})_{123,123} = G_{44}/\det G.$$

These complementary-minor formulas follow from the cofactor inverse formula for three-element minors. For the two-element formula in FC60, block-eliminate the positive principal block $G_{34,34}$. The complementary block of G^{-1} is the inverse Schur complement, and $\det G$ is the product of that complement's determinant with $\det G_{34,34}$, giving the claimed identity without discarding any cross term.

Continuity of the norm of a finite matrix with fixed positive domain and codomain forms now proves

r	$\lim_{\epsilon \downarrow 0} \epsilon^{m_r} \ \wedge^r DP_{\mathcal{W}}(z_\epsilon)\ $	$\lim_{\epsilon \downarrow 0} \epsilon^{m_r} \ \wedge^r DP_{\mathcal{W}}(z_\epsilon)^{-1}\ $	
1	$64\tau_1$	$120\tau_1$	
2	$672\tau_2$	$336\tau_2$	
3	$240\tau_1$	$32\tau_1$	
4	2	$1/2$	.

(FC61)

Every norm here is in the original target Gamma metric and its induced exterior metric. The pulled-back matrix G is only its exact coordinate expression; the maps Ψ and K have not been declared unitary.

Let $\sigma_1(\epsilon) \geq \dots \geq \sigma_4(\epsilon) > 0$ be the four singular values of this derivative in that metric. The largest singular value of $\wedge^r J$ is the product of its largest r singular values: diagonalizing the two positive singular-value factors gives the wedge basis with all r -fold products. Taking successive ratios of FC61 therefore proves the full spectrum

$$\begin{aligned} \sigma_1(\epsilon) &\sim 64\tau_1\epsilon^{-4}, & \sigma_2(\epsilon) &\sim \frac{21\tau_2}{2\tau_1}\epsilon^{-1}, \\ \sigma_3(\epsilon) &\sim \frac{5\tau_1}{14\tau_2}\epsilon, & \sigma_4(\epsilon) &\sim \frac{1}{120\tau_1}\epsilon^4. \end{aligned} \tag{FC62}$$

The product of the four displayed constants is 2, agreeing with the exact determinant modulus. In particular the condition number is asymptotic to $7680\tau_1^2\epsilon^{-8}$. The degeneration is at infinity along a proved branch; each finite point still has nonzero Jacobian determinant.

There is also an exact inverse derivative identity in the direction of the image curve. Differentiating FC45–46 gives

$$J_\epsilon \begin{pmatrix} -1 \\ -2i\epsilon^{-2} \\ -12i\epsilon^{-3} \\ -120i\epsilon^{-4} \end{pmatrix} = \begin{pmatrix} -i + 3i\epsilon^2/2 \\ 3i/2 \\ 0 \\ 0 \end{pmatrix}. \tag{FC63}$$

This identifies a specific bounded target velocity whose inverse grows as ϵ^{-4} , in addition to the full singular spectrum. If $p_\epsilon = P_{\mathcal{W}}(z_\epsilon)$, put $v_0 = \Psi(-i, 3i/2, 0, 0)^T \neq 0$ and $p_* = 2iy_M$. FC46 implies $(p_\epsilon - p_*)/\epsilon \rightarrow v_0$. Combining this limit with FC48 and FC61 gives the exact distance laws

$$\|p_\epsilon - p_*\|^3 \|z_\epsilon\| \rightarrow 40 \|\Psi e_w\| \|v_0\|^3, \quad \|p_\epsilon - p_*\|^4 \|DP_{\mathcal{W}}(z_\epsilon)^{-1}\| \rightarrow 120\tau_1 \|v_0\|^4. \tag{FC64}$$

All norms and all constants in these two limits are the same original Gamma norms of FC56. Injectivity of Ψ proves their positivity.

12 The exact return to audit equation (11)

For an invertible map B of a four-dimensional Hilbert space, its singular values prove

$$\|\wedge^r B^{-1}\| = \frac{\|\wedge^{4-r} B\|}{|\det B|}. \quad (\text{FC65})$$

Applying this to FC57 gives precisely the inverse constants in FC61, including $\|\wedge^2 B^{-1}\| = \|\wedge^2 B\|/2$. Thus the escaping branch realizes highly anisotropic distortion while satisfying the same exterior identity used to derive audit (11). Its forward bound is not fixed along the branch: it grows with the explicit poles in FC59. No contradiction with that identity occurs.

The exact relation to the original conductor is stronger than applying the identity to an unrelated matrix. Put $L = T_A|_{\mathcal{U}}$, $u_\epsilon = \Phi\hat{q}_\epsilon$, and $B_{U,\epsilon} = DP_{\mathcal{U}}(u_\epsilon)$, $B_{W,\epsilon} = DP_{\mathcal{W}}(z_\epsilon)$. Differentiating FC33 at the very same source point proves

$$LB_{U,\epsilon} = B_{W,\epsilon}L, \quad B_{W,\epsilon}^{-1} = LB_{U,\epsilon}^{-1}L^{-1}. \quad (\text{FC66})$$

This identity retains the original source quotient metric on $\mathcal{U}$ and target metric on $\mathcal{W}$. In particular, for every $0 \leq r \leq 4$,

$$\frac{\|\wedge^r B_{U,\epsilon}^{-1}\|}{\|\wedge^r L\| \|\wedge^r L^{-1}\|} \leq \|\wedge^r B_{W,\epsilon}^{-1}\| \leq \|\wedge^r L\| \|\wedge^r L^{-1}\| \|\wedge^r B_{U,\epsilon}^{-1}\|. \quad (\text{FC67})$$

The upper bound follows from composition of exterior maps; the lower bound follows by solving FC66 for $B_{U,\epsilon}^{-1}$. On the four-plane, L is exactly $A_{3,s}$ in the WCF orthonormal frames, so the actual bound in audit (11) applies to the unchanged factor $\|\wedge^r L^{-1}\|$. For fixed conductor this factor is independent of ϵ . All four pole orders therefore survive transport through the original conductor in both directions. This supplies a quantitative return to the audited map, without assigning to it the nonlinear map's variable derivative or removing a kernel without its quotient.

For completeness, define the source pullback form $G_U = (K^{-1})^*G_xK^{-1}$. Replacing G in FC60 by G_U proves the exact source constants in FC61–64 as well, since the Laurent matrices FC57–59 are unchanged. Both source and target degenerations are now evaluated in their original norms. The constant determinant and the nonproper inverse are compatible; the first does not control the separate singular values, which FC62 evaluates completely.

13 The singular flags and the exact volumes of fixed Fable flags

The compound limits determine the directions of this distortion as well as its size. Let $I_k = \{1, \dots, k\}$ and $J_k = \{5 - k, \dots, 4\}$. For $k = 1, 2, 3$, the limiting right singular k -plane in the G coordinates and the limiting left plane are

$$\mathcal{R}_k = G^{-1} \text{span}\{e_j : j \in I_k\}, \quad \mathcal{L}_k = \text{span}\{e_j : j \in J_k\}. \quad (\text{FC68})$$

To prove this, the rank-one compound limit in FC59 has image line $\mathbb{C}e_{J_k}$ and covector proportional to $e_{I_k}^*$. The Riesz vector of the coordinate covector e_j^* in the original form is $G^{-1}e_j$, because $\langle G^{-1}e_j, x \rangle_G = e_j^*x$. Taking wedges gives the right singular line $\mathbb{C}\bigwedge_{j \in I_k}(G^{-1}e_j)$. The nonzero singular value of each limiting rank-one map is simple; its other singular values are zero. For convergent matrices with this property, their leading left and right singular lines converge: a convergent subsequence of unit maximizing vectors must attain the limiting positive norm and therefore belongs to that unique line. Compactness of the unit sphere and uniqueness give convergence of the full family of lines. The Plücker line of a decomposable nonzero wedge determines its plane by $x \wedge \omega = 0$, so the singular k -planes have the stated limits. Applying Ψ gives these planes in the literal original target, with its unchanged Gamma metric.

In particular, the greatest right-singular direction tends to $\mathbb{C}G^{-1}e_1$, and the greatest left-singular direction tends to $\mathbb{C}e_4$. Applying the inverse compound limit in FC59 shows that the least right-singular direction tends to $\mathbb{C}e_4$, and its least left-singular direction tends to $\mathbb{C}G^{-1}e_1$. These are directions, not a declaration that the coordinate vectors are orthogonal or of unit length.

There is also a complete calculation for fixed, explicitly transported subspaces, as in the audit's endpoint Gram formula (12). Let

$$E_k = \Psi \operatorname{span}\{e_1, \dots, e_k\} \subset \mathcal{W}, \quad 1 \leq k \leq 4, \quad (d_1, d_2, d_3, d_4) = (64, 672, 240, 2).$$

The ratio of the output Gram determinant of this particular frame under $B_{W,\epsilon}$ to its input Gram determinant is exactly

$$\mathcal{V}_k(\epsilon) = \frac{\det((J_\epsilon^* G J_\epsilon)_{I_k, I_k})}{\det G_{I_k, I_k}}, \quad \epsilon^{2m_k} \mathcal{V}_k(\epsilon) \longrightarrow d_k^2 \frac{\det G_{J_k, J_k}}{\det G_{I_k, I_k}} > 0. \quad (\text{FC69})$$

Indeed the numerator is the squared original wedge norm of $(\wedge^k J_\epsilon)e_{I_k}$. FC59 takes this wedge, after the displayed power of ϵ , to a scalar of modulus d_k times e_{J_k} . Taking its squared norm proves FC69, while the denominator is the exact squared norm of the input wedge. Hence

$$\log \mathcal{V}_k(\epsilon) = 2m_k \log(1/\epsilon) + \log \left(d_k^2 \frac{\det G_{J_k, J_k}}{\det G_{I_k, I_k}} \right) + o(1). \quad (\text{FC70})$$

For $k = 4$, this ratio is identically 4, not merely asymptotic to it. For $k = 1, 2, 3$ it diverges with the specified rates, even though the full determinant stays fixed. This computes actual subspaces and their exact original metric constants; it is not an upper enclosure for an unspecified endpoint. The E_k are the transported Fable coordinate flags defined here, and no equality with a different programme endpoint is inferred. Their source partners $\Phi \operatorname{span}(e_1, \dots, e_k)$ are carried to them by the original L in FC66. Replacing G by G_U gives their full source volume formulas by the same proof.

The difference between the actual coordinate-flag constant and the optimizing singular-flag constant is itself exact. For $k = 1, 2, 3$, FC59–60 and FC69 give

$$\lim_{\epsilon \downarrow 0} \frac{\|\wedge^k B_{W,\epsilon}\|^2}{\mathcal{V}_k(\epsilon)} = \det G_{I_k, I_k} \det(G^{-1})_{I_k, I_k} \geq 1, \quad (\text{FC71})$$

with equality precisely when $G_{I_k, I_k^c} = 0$. To prove this without omitting the cross terms, write G in these blocks as $\begin{pmatrix} A & C \\ C^* & D \end{pmatrix}$. The corresponding inverse block is $(A - CD^{-1}C^*)^{-1}$. The matrix $A^{-1/2}(A - CD^{-1}C^*)A^{-1/2}$ is positive and at most the identity, since $CD^{-1}C^*$ is positive semidefinite. Its determinant is at most one, which proves the inequality. Equality makes all its eigenvalues one, hence $CD^{-1}C^* = 0$. Positive definiteness of D^{-1} then gives $C = 0$; the reverse implication follows by substitution. The limit equality follows by dividing the rank-one operator constant $d_k^2 \det G_{J_k, J_k} \det(G^{-1})_{I_k, I_k}$ by the fixed-frame constant in FC69. Thus the exact preserved cross Gram determines the slack in replacing these particular flag volumes by the operator bound used in audit (11).

The following complete inverse calculation extends the canonical ES source [1], whose original polynomial and four marked representatives are retained. It classifies every target, then specializes the result to the literal four roots used by the original conductor.

14 The original polynomial and the factor coordinates

Fix $i^2 = -1$. The original coordinates are $(a, y, z, w) \in \mathbb{C}^4$, and the ordered output coordinates are (u_0, u_2, u_3, u_4) . Define

$$\begin{aligned} P_0 &= a^3z + 2a^2y - ia, \\ P_2 &= -a^3y^2z - 2ia^2yz + a^2w - 2a^2y^3 - 10ia y^2 + 3az + y, \\ P_3 &= 2a^3y^3z + 6ia^2y^2z + 2a^2wy + 4a^2y^4 + 2ia w - 4ia y^3 - 2ayz + 2iz + 7y^2, \\ P_4 &= 2a^3y^4z + 8ia^2y^3z + a^2wy^2 + 4a^2y^5 + 2ia wy + 7ia y^4 - 10ay^2z - 4iyz - w - 3y^3. \end{aligned} \quad (\text{GF1})$$

The factor-coordinate map $\Phi_{\text{fac}}(a, y, z, w) = (a, b, c, d, e, f)$ is

$$\begin{aligned} b &= i + ay, & c &= -i + 2ay + a^2z, \\ d &= -iy - a(iz + 2y^2) - a^2yz, & e &= 2z - 7iy^2 + aw, \\ f &= iw + 3iy^3 - 4yz + a(6iy^2z + wy + 4y^4) + 2a^2y^3z. \end{aligned} \quad (\text{GF2})$$

Let

$$L = aU + bV, \quad C = cU^3 + dU^2V + eUV^2 + fV^3.$$

Their original incidence and resultant equations are

$$ad + bc = 1, \quad R(L, C) = C(-b, a) = a^3f - a^2be + ab^2d - b^3c = 1. \quad (\text{GF3})$$

Direct substitution of GF2 proves both identities. Multiplication of the two factors gives the complete coefficient map

$$LC = (ac)U^4 + U^3V + (ae + bd)U^2V^2 + (af + be)UV^3 + (bf)V^4, \quad P = (ac, ae + bd, af + be, bf). \quad (\text{GF4})$$

Substitution of GF2 in GF4 gives precisely GF1.

On $a \neq 0$, every factor state satisfying GF3 has the unique inverse

$$y = (b - i)/a, \quad z = (c + i - 2ay)/a^2, \quad w = (e - 2z + 7iy^2)/a. \quad (\text{GF5})$$

These formulas recover a, b, c, e . The first equation GF3 uniquely determines d , and its second uniquely determines f because $a^3 \neq 0$. Thus GF5 is a full inverse on that open set. At $a = 0$, GF3 forces $bc = 1$ and $-b^3c = 1$, hence $b = \pm i$. The original chart GF2 has $b = i$, $c = -i$, and contains all such states: its inverse there is

$$y = id, \quad z = (e + 7iy^2)/2, \quad w = (f - 3iy^3 + 4yz)/i.$$

It contains none of the $a = 0, b = -i$ states. This description of the chart is used explicitly in every fibre count below.

15 Exact inverse over every target

For $u = (u_0, u_2, u_3, u_4)$ define

$$H_u(U, V) = u_0U^4 + U^3V + u_2U^2V^2 + u_3UV^3 + u_4V^4, \quad h_u(x) = H_u(x, 1). \quad (\text{GF6})$$

The fixed U^3V coefficient is part of the original map, not a choice made during this calculation.

Theorem 1 (Complete fibre, including every multiple-root case). *Every finite simple root r of h_u contributes exactly two distinct points of $P^{-1}(u)$. Their complete coordinates are*

$$\begin{aligned} a^2 &= 1/h'_u(r), & a &\in \{\text{both square roots}\}, \\ y &= -r - i/a, \\ z &= u_0/a^3 + 2r/a + 3i/a^2, \\ w &= 7ir^2/a + (u_2 - 17r + u_0r^2)/a^2 - 13i/a^3 - 2u_0/a^4. \end{aligned} \quad (\text{GF7})$$

A finite multiple root contributes no point. When $u_0 = 0$ there is exactly one additional point, namely

$$q_\infty(u) = \left(0, u_2, \frac{i(7u_2^2 - u_3)}{2}, 11u_2^3 - 2u_2u_3 - u_4\right). \quad (\text{GF8})$$

There are no other points. In particular, writing $m(u)$ for the number of distinct finite simple roots,

$$\#P^{-1}(u) = 2m(u) + \mathbf{1}_{\{u_0=0\}}. \quad (\text{GF9})$$

Proof. A source point with $a \neq 0$ gives the marked finite root $r = -b/a$ of $H_u = LC$. The quotient factor is necessarily

$$\begin{aligned} b &= -ar, & c &= u_0/a, & d &= (1 + u_0r)/a, \\ e &= (u_2 + r + u_0r^2)/a, & f &= (u_3 + u_2r + r^2 + u_0r^3)/a. \end{aligned}$$

These formulas follow by comparing successive coefficients in $H_u = a(U - rV)C$. The final constant coefficient is equivalent to $h_u(r) = 0$. Evaluating C at $(-b, a) = (ar, a)$ gives

$$R(L, C) = a^3C(r, 1) = a^2h'_u(r).$$

Thus GF3 holds exactly when $a^2h'_u(r) = 1$. A multiple root is excluded by this equation, while each simple root supplies exactly two nonzero values of a . The other incidence equation follows without a further constraint: $ad + bc = 1 + u_0r - u_0r = 1$. The complete inverse GF5 now gives GF7; for example substitution in $w = (e - 2z + 7iy^2)/a$ yields all four displayed terms of its last line. These operations are reversible, so they prove both sufficiency and exhaustiveness on $a \neq 0$. Distinct roots give different $r = -b/a$, and the two signs of a give distinct source points.

If $a = 0$ then $u_0 = ac = 0$. Conversely, GF1 on this hyperplane reads

$$P(0, y, z, w) = (0, y, 2iz + 7y^2, -4iyz - w - 3y^3).$$

Solving successively gives exactly GF8. In projective factor terms, $u_0 = 0$ makes $[U : V] = [1 : 0]$ a simple root: $H_u = V(U^3 + \dots)$ and the leading coefficient of the parenthesis is exactly one. Its two resultant-one factor scalings have $b = \pm i$, but GF2 retains only $b = i$. This is precisely the single point just solved, regardless of any multiple finite roots. Hence there are no omitted chart exceptions, and GF9 follows. $\square$

For clarity the exhaustive multiplicity counts are

target chart	finite root multiplicities	$\#P^{-1}(u)$
$u_0 \neq 0$	$1 + 1 + 1 + 1$	8
$u_0 \neq 0$	$2 + 1 + 1$	4
$u_0 \neq 0$	$3 + 1$	2
$u_0 \neq 0$	$2 + 2$ or 4	0
$u_0 = 0$	$1 + 1 + 1$	7
$u_0 = 0$	$2 + 1$	3
$u_0 = 0$	3	1

Every row follows directly from GF9. The polynomial always has degree four or three, because the cubic coefficient is fixed at one.

16 A short complete Jacobian proof

On $a \neq 0$, use parameters (a, r, A, B) , where $A = u_0$ and $B = u_2$. The two equations $h_u(r) = 0$, $a^2 h'_u(r) = 1$ solve as

$$u_3 = a^{-2} - 4Ar^3 - 3r^2 - 2Br, \quad u_4 = 3Ar^4 + 2r^3 + Br^2 - r/a^2. \quad (\text{GF10})$$

GF7 gives the source parameterization

$$q(a, r, A, B) = (a, -r - i/a, A/a^3 + 2r/a + 3i/a^2, \\ 7ir^2/a + (B - 17r + Ar^2)/a^2 - 13i/a^3 - 2A/a^4).$$

Its Jacobian determinant in this parameter order is $-a^{-5}$: after its first row, the remaining triangular minor has diagonal $-1, a^{-3}, a^{-2}$. For the target parameterization (A, B, u_3, u_4) , re-order the first two parameter columns past the last two; this permutation has positive sign. The remaining two-by-two determinant is

$$\det \begin{pmatrix} -2/a^3 & -(12Ar^2 + 6r + 2B) \\ 2r/a^3 & r(12Ar^2 + 6r + 2B) - a^{-2} \end{pmatrix} = 2a^{-5}.$$

The chain rule therefore proves

$$\det DP = -2 \quad \text{on } a \neq 0. \quad (\text{GF11})$$

Every original source point with $a \neq 0$ has these parameters by GF5–7, so this identity holds on that entire open set. Both sides of GF11 are polynomials in the original coordinates; their difference vanishes on a nonempty dense open set and is consequently the zero polynomial. This proves GF11 everywhere, including the $a = 0$ chart, with its original constant and sign.

17 The exact omitted image

The only quartic multiplicity patterns with no simple root are $2 + 2$ and 4 . Equivalently a quartic with $u_0 \neq 0$ and no simple root is the square of a quadratic times its leading coefficient. Because its cubic coefficient is one, it must have the form

$$H_u(U, V) = u_0 \left(U^2 + \frac{1}{2u_0} UV + u_3 V^2 \right)^2.$$

Indeed, write the monic quadratic as $U^2 + pUV + qV^2$; comparison of cubic and linear coefficients gives $p = 1/(2u_0)$ and $q = u_3$. The remaining coefficients then give the exact image complement

$$\mathbb{C}^4 \setminus P(\mathbb{C}^4) = \{4u_0u_2 - 1 - 8u_0^2u_3 = 0, \quad u_4 - u_0u_3^2 = 0\}. \quad (\text{GF12})$$

The first equation excludes $u_0 = 0$. Conversely the two equations give exactly the displayed square. GF9 therefore proves both inclusions in GF12, not only exclusion of the square targets.

18 Discriminant and the genuine eight-sheet domain

Write $A = u_0, B = u_2, C = u_3, D = u_4$. Define the original binary-quartic discriminant polynomial by

$$\mathfrak{D}(u) = 256A^3D^3 - 128A^2B^2D^2 + 144A^2BC^2D - 27A^2C^4 - 192A^2CD^2 \\ + 16AB^4D - 4AB^3C^2 - 80AB^2CD + 18ABC^3 + 144ABD^2 - 6AC^2D \\ - 4B^3D + B^2C^2 + 18BCD - 4C^3 - 27D^2. \quad (\text{GF13})$$

For $A \neq 0$, the Sylvester determinant of h_u and h'_u , divided by A , expands to GF13. If $r_1, \dots, r_4$ are its roots counted with multiplicity, that same determinant gives

$$\mathfrak{D} = A^6 \prod_{j < k} (r_j - r_k)^2. \quad (\text{GF14})$$

One can verify this identity first for distinct roots: the resultant is $A^3 \prod_j h'_u(r_j)$, and $\prod_j h'_u(r_j) = A^4 \prod_{j < k} (r_j - r_k)^2$ because $(-1)^{4 \cdot 3/2} = 1$. Polynomial continuity gives the identity also at multiple roots. At $A = 0$, GF13 becomes

$$B^2 C^2 - 4C^3 - 4B^3 D + 18BCD - 27D^2,$$

the discriminant of the monic cubic $x^3 + Bx^2 + Cx + D$. The same resultant calculation in degree three proves that it vanishes exactly at a repeated finite root. Since the projective root at infinity is simple when $A = 0$, GF13 vanishes exactly when H_u has any multiple projective root, on both charts.

Let

$$\mathcal{B} = \{u \in \mathbb{C}^4 : u_0 \mathfrak{D}(u) \neq 0\}, \quad R = \mathbb{C}[u_0, u_2, u_3, u_4, (u_0 \mathfrak{D})^{-1}]. \quad (\text{GF15})$$

Over this precise domain the map P is an eight-sheet finite unramified cover. Its complete coordinate algebra is

$$S = R[r, a]/(h_u(r), a^2 h'_u(r) - 1). \quad (\text{GF16})$$

Here is a direct proof of each assertion. Since u_0 is a unit, the first relation is a monic relation of degree four after division by u_0 , and $R[r]/(h_u)$ is free of rank four over R with basis $1, r, r^2, r^3$. The resultant of h_u and h'_u is a unit in R . The adjugate of their Sylvester matrix therefore gives a Bezout identity over R , so $h'_u(r)$ is a unit in this rank-four algebra. The second relation is consequently the monic relation $a^2 = h'_u(r)^{-1}$. Thus S is free of rank eight over R , with basis $1, r, r^2, r^3, a, ar, ar^2, ar^3$.

The original chart inverse GF7 has entries in S : a is a unit with inverse $ah'_u(r)$. In the other direction, for an original source point over $\mathcal{B}$ the identity $u_0 = ac$ makes a a unit, and $r = -b/a$ is regular. GF3–7 prove these two maps are mutually inverse. Thus GF16 is the coordinate algebra of the *actual* preimage $P^{-1}(\mathcal{B})$, rather than a new eight-point model.

The derivative matrix of the two defining equations of GF16 with respect to (r, a) has determinant $2a h'_u(r)^2$, a unit. This proves that each of its eight simple solutions gives one local holomorphic inverse, equivalently that the finite free map is unramified. Explicitly the four simple roots vary holomorphically in a sufficiently small target neighbourhood, and each nonzero $1/h'_u(r)$ has its two separate holomorphic square roots there. GF7 gives eight disjoint local inverse charts. There are exactly eight by GF9. These statements require neither a chosen global square-root branch nor identification of its two signs.

19 The complete nonproper-value locus

A target u is called a nonproper value here if there is a sequence $q_n \in \mathbb{C}^4$ escaping every compact set with $P(q_n) \rightarrow u$. For this polynomial map that is the local failure of properness near u . The exact locus is

$$\mathcal{N}(P) = \{u_0 = 0\} \cup \{\mathfrak{D}(u) = 0\}. \quad (\text{GF17})$$

First take $u \in \mathcal{B}$ and a convergent sequence of targets $u_n \rightarrow u$. On a sufficiently small closed neighbourhood of u , u_0 and $\mathfrak{D}$ are bounded away from zero. Every finite root is bounded: after dividing h by u_0 , any root has modulus at most one plus the maximum modulus of its nonleading coefficients. This follows by comparing the leading term with the finite geometric sum of the remaining terms outside that disk. Moreover $|h'_{u_n}(r_n)|$ is bounded away from zero uniformly over all roots. Otherwise a convergent subsequence of the bounded r_n would give a root r of h_u

with $h'_u(r) = 0$, contradicting $\mathfrak{D}(u) \neq 0$. The derivatives also have a uniform upper bound, by the root and coefficient bounds. Hence $a^2 = 1/h'(r)$ bounds both a and a^{-1} ; GF7 then bounds every original source coordinate. There is no $a = 0$ solution on this neighbourhood. Thus no sequence above u_n can escape, proving $\mathcal{N}(P) \subseteq \mathbb{C}^4 \setminus \mathcal{B}$.

For the reverse inclusion at $u_0 = 0$, let GF8 be $q_0 = (0, y_0, z_0, w_0)$ and set $q_\epsilon = (\epsilon, y_0, z_0, w_0)$ for real $\epsilon > 0$. On $a \neq 0$ the exact factor-sign involution is

$$\Sigma(a, y, z, w) = \left(-a, y + \frac{2i}{a}, \frac{6i}{a^2} - z, w - \frac{14iy^2}{a} + \frac{28y}{a^2} + \frac{40i}{a^3} \right). \quad (\text{GF18})$$

To verify its invariance, substitute GF18 in GF2: a, b, c, e all change sign. GF3 then forces d and f also to change sign because $a \neq 0$. Both factors in GF4 change sign, so $P\Sigma = P$. The full inverse GF5 proves $\Sigma^2 = I$ as well. Now the second coordinate of Σq_ϵ is $y_0 + 2i/\epsilon$ and escapes, while $P\Sigma q_\epsilon = Pq_\epsilon \rightarrow Pq_0 = u$. Every point of the entire hyperplane $u_0 = 0$ is consequently nonproper, including its discriminant intersections.

Finally let $u_0 \neq 0$ and $\mathfrak{D}(u) = 0$. Choose any multiple root r_0 of h_u and define

$$h_\epsilon(x) = h_u(x) + \epsilon(x - r_0), \quad u_\epsilon = (u_0, u_2, u_3 + \epsilon, u_4 - \epsilon r_0).$$

These preserve the original fixed cubic coefficient. The root r_0 persists and has derivative exactly ϵ . For real $\epsilon > 0$ choose $a = \epsilon^{-1/2}$ and apply the complete inverse GF7 to this simple root. It gives a genuine source point q_ϵ with $Pq_\epsilon = u_\epsilon \rightarrow u$, while its first coordinate tends to infinity. This argument works for every multiplicity of r_0 and does not require the other roots to be simple. It proves the remaining inclusion in GF17.

20 The literal arithmetic quartet as a marked target quartic

Retain the original four distinct roots, with both original signs:

$$\rho_{\epsilon, \eta} = \frac{1}{2} + \epsilon\delta + i\eta\gamma, \quad \epsilon, \eta \in \{1, -1\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2.$$

Their sum is exactly two. Therefore the quartic with these literal roots and the required original cubic coefficient one is

$$H_\rho(U, V) = -\frac{1}{2} \prod_{\epsilon, \eta} (U - \rho_{\epsilon, \eta} V). \quad (\text{GF19})$$

The factor $-1/2$ is forced by the cubic coefficient and is retained in every derivative, discriminant, and inverse below. With $x = S - 1/2$, its affine polynomial is

$$h_\rho(S) = -\frac{1}{2}(x^4 + 2(\gamma^2 - \delta^2)x^2 + (\delta^2 + \gamma^2)^2).$$

Expanding gives the exact target

$$u_\rho(\delta) = \left(-\frac{1}{2}, \delta^2 - \gamma^2 - \frac{3}{4}, \gamma^2 - \delta^2 + \frac{1}{4}, -\frac{1}{32} - \frac{\gamma^2 - \delta^2}{4} - \frac{(\delta^2 + \gamma^2)^2}{2} \right). \quad (\text{GF20})$$

At the marked root $\rho_{\epsilon, \eta}$,

$$h'_\rho(\rho_{\epsilon, \eta}) = 4\delta\gamma(\epsilon\gamma - i\eta\delta), \quad a_{\epsilon, \eta, \pm}^2 = \frac{1}{4\delta\gamma(\epsilon\gamma - i\eta\delta)}. \quad (\text{GF21})$$

To verify the derivative, the three differences from that root are $2i\eta\gamma$, $2\epsilon\delta$, and $2\epsilon\delta + 2i\eta\gamma$; multiply them and the retained leading coefficient $-1/2$. This gives precisely GF21. Both square roots are present at every one of the four marked roots.

In the order $(+, +), (+, -), (-, +), (-, -)$ the original Vandermonde is $\Delta_\rho = -64\delta^2\gamma^2(\delta^2 + \gamma^2)$. Hence

$$\mathfrak{D}(u_\rho) = \Delta_\rho^2/64 = 64\delta^4\gamma^4(\delta^2 + \gamma^2)^2 > 0. \quad (\text{GF22})$$

The denominator 64 is the sixth power of the retained factor 2. Thus $u_\rho \in \mathcal{B}$, and GF7 and GF21 give all eight original chart states over it. Unlike the earlier target $UV(U - V)(U + V)$, it has no root at infinity and loses no sign partner from the source chart.

For fixed γ and $\delta \downarrow 0$, GF19 tends to

$$H_0 = -\frac{1}{2} \left((U - V/2)^2 + \gamma^2 V^2 \right)^2. \quad (\text{GF23})$$

This target is on the omitted square surface GF12, so its complete source fibre is empty. The eight nearby states all escape; they do not become fewer finite chart points. Indeed GF21 gives

$$|a| = (4\delta\gamma\sqrt{\gamma^2 + \delta^2})^{-1/2}, \quad \sqrt{\delta}|a| \longrightarrow \frac{1}{2\gamma}.$$

For each continuous choice of its sign branch,

$$\sqrt{\delta} a_{\varepsilon, \eta, \pm} \longrightarrow \kappa_{\varepsilon, \pm}, \quad \kappa_{\varepsilon, \pm}^2 = \frac{1}{4\varepsilon\gamma^2}. \quad (\text{GF24})$$

GF7 then gives $y \rightarrow -(1/2 + i\eta\gamma)$ and $z, w \rightarrow 0$. Consequently, in the original source coordinates,

$$\sqrt{\delta} q_{\varepsilon, \eta, \pm}(\delta) \longrightarrow \kappa_{\varepsilon, \pm} e_a, \quad e_a = (1, 0, 0, 0)^\top. \quad (\text{GF25})$$

Each claim follows from the displayed inverse formulas; no moving period or limiting identification of distinct root coordinates is used to infer these limits.

21 All eight escaping branches in one fixed original conductor norm

Fix once and for all an admitted reference quartet with parameter $\delta_* > 0$, its original conductor, an original permitted source order s , and its original target Gamma norm on $W = \mathcal{P}_3$. Let $\Psi_* : \mathbb{C}^4 \rightarrow W$ be the exact invertible receiving map constructed from that reference conductor in FC26–35 above. Its entries include the reference period and every reference moment; none is varied in this section. Define

$$K = \begin{pmatrix} 0 & i & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1 & -1 - \sqrt{2}i & 1 - \sqrt{2}i \\ i/2 & -3i & 2\sqrt{2} + 6i & -2\sqrt{2} + 6i \\ 0 & 13 & -34 - 19\sqrt{2}i & 34 - 19\sqrt{2}i \end{pmatrix}, \quad \det K = 77\sqrt{2}i/2.$$

For complete coordinates, let $V_* = (\rho_{*,j}^n)_{0 \leq n \leq 3, 1 \leq j \leq 4}$ be the reference root Vandermonde, and let μ_j and v be the original reference conductor moments and their actual first nonzero order. In the target basis $(1, S', S'^2, S'^3)$ the complete matrix of this fixed map is

$$[\Psi_*] = B_{v,*}(V_*^\top)^{-1}K^{-1}, \quad (B_{v,*})_{kj} = \begin{cases} \binom{v+j}{k} \mu_{v+j-k}, & 0 \leq k \leq j \leq 3, \\ 0, & k > j. \end{cases}$$

The first determinant is $\det B_{v,*} = \mu_v^4 \prod_{j=0}^3 \binom{v+j}{v} \neq 0$; the reference Vandermonde is nonzero and K has the determinant displayed above. This proves the asserted invertibility with all original factors retained. The target norm in this monomial frame is the original one, not the Euclidean norm of the column:

$$\|p\|_{\Gamma,*}^2 = \int_{\mathbb{R}} |p(c_* + iy)|^2 \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad c_* = a_{\text{amp},*}/2 - 4,$$

where $s \in \{1, a_{\text{amp},*}\}$ is the fixed original reference source order. This integral defines a positive definite norm on $\mathcal{P}_3$: its density is positive on the real axis and a nonzero polynomial cannot vanish on that entire axis. Finiteness is the original WCF2 orthogonal-polynomial norm identity $h_{n,s} = (2\pi)^{s/2} n! (s/2)_n$, applied to the finite expansion of the polynomial in that frame, with no change of measure. With these exact maps and norms, define

$$P_{W,*} = \Psi_* P \Psi_*^{-1}, \quad p_\delta = \Psi_* u_\rho(\delta), \quad Z_{\varepsilon,\eta,\pm}(\delta) = \Psi_* q_{\varepsilon,\eta,\pm}(\delta).$$

The variable δ in these formulas parametrizes an auxiliary target path inside this *fixed* operator and its *fixed* norm. It does not move the reference arithmetic period. Exact conjugacy gives all eight fibre identities

$$P_{W,*}(Z_{\varepsilon,\eta,\pm}(\delta)) = p_\delta.$$

GF25 and continuity of the fixed linear map give the actual norm asymptote, for every one of the eight branches,

$$\sqrt{\delta} \|Z_{\varepsilon,\eta,\pm}(\delta)\|_{\Gamma,*} \longrightarrow \frac{\|\Psi_* e_a\|_{\Gamma,*}}{2\gamma} > 0. \quad (\text{GF26})$$

The strict inequality follows because Ψ_* is injective and the original Gamma norm is positive definite.

The target displacement in its original coordinates is exactly

$$u_\rho(\delta) - u_\rho(0) = \delta^2 d_\gamma - \frac{1}{2} \delta^4 e_w, \quad d_\gamma = (0, 1, -1, \frac{1}{4} - \gamma^2)^\top, \quad e_w = (0, 0, 0, 1)^\top. \quad (\text{GF27})$$

Subtract GF20 at zero to verify every coefficient. Therefore $\delta^{-2} \|p_\delta - p_0\|_{\Gamma,*} \rightarrow \|\Psi_* d_\gamma\|_{\Gamma,*} > 0$. Combining this with GF26 proves the complete original-norm rate

$$\boxed{\|p_\delta - p_0\|_{\Gamma,*}^{1/4} \|Z_{\varepsilon,\eta,\pm}(\delta)\|_{\Gamma,*} \longrightarrow \frac{\|\Psi_* d_\gamma\|_{\Gamma,*}^{1/4} \|\Psi_* e_a\|_{\Gamma,*}}{2\gamma} > 0.} \quad (\text{GF28})$$

The limiting point p_0 has no finite preimage under $P_{W,*}$, because GF12 and the exact conjugacy transfer that assertion as well. For completeness, the entire nonproper-value locus of this fixed nonlinear receiving map is precisely $\Psi_*(\{u_0 \mathfrak{D}(u) = 0\})$: invertible linear maps preserve boundedness, escaping sequences, and convergence. This statement concerns $P_{W,*}$; the original linear conductor remains the invertible intertwiner defining Ψ_* .

GF26–28 avoid using a singular limit of a moving Vandermonde or a moving conductor norm. Such a limit would be a separate calculation: the construction of Ψ_δ itself assumes distinct original roots, and its original Vandermonde vanishes at $\delta = 0$. Everything asserted here has instead been proved with the reference operator, its full original coefficients, and its original norm held fixed. No assertion about RH follows from introducing this nonlinear map or this auxiliary target path.

22 The eight-state evaluation and its exact sign decomposition

The eight inverse states from GF7 give a second, intrinsic four-vector frame in the original conductor target. This construction uses the actual opposite-sign pairs in that fibre, rather than selecting four unrelated vectors. Write $A = u_0$, $B = u_2$, $C = u_3$, $D = u_4$ and $h(r) = Ar^4 + r^3 + Br^2 + Cr + D$, retaining $A \neq 0$. For each of its four distinct roots r_j choose one of the two values $a_j^2 h'(r_j) = 1$, and write $q_{j,+} = q(r_j, a_j)$ and $q_{j,-} = q(r_j, -a_j)$ using the full inverse

GF7. Put $d_j = h'(r_j)$. Direct addition and subtraction of GF7 give

$$\begin{aligned} e_j &:= \frac{q_{j,+} + q_{j,-}}{2} = \begin{pmatrix} 0 \\ -r_j \\ 3id_j \\ (B - 17r_j + Ar_j^2)d_j - 2Ad_j^2 \end{pmatrix}, \\ o_j &:= \frac{q_{j,+} - q_{j,-}}{2} = a_j \begin{pmatrix} 1 \\ -id_j \\ Ad_j^2 + 2r_jd_j \\ 7ir_j^2d_j - 13id_j^2 \end{pmatrix}. \end{aligned} \tag{SE1}$$

For example $a_j^{-1} = a_j d_j$, $a_j^{-2} = d_j$, $a_j^{-3} = a_j d_j^2$, and $a_j^{-4} = d_j^2$; these identities retain every constant and sign in that addition. Thus SE1 is an exact reconstruction $q_{j,\pm} = e_j \pm o_j$.

Let $\mathcal{F}_u = \mathbb{C}\{(j,+), (j,-): 1 \leq j \leq 4\}$ with its eight separately marked basis vectors $f_{j,\pm}$, and let $Sf_{j,\pm} = f_{j,\mp}$. The two explicit embeddings are

$$\iota_+ e_j^{\text{coord}} = (f_{j,+} + f_{j,-})/2, \quad \iota_- e_j^{\text{coord}} = (f_{j,+} - f_{j,-})/2. \tag{SE2}$$

Their images are the $+1$ and -1 eigenspaces of S ; every vector of $\mathcal{F}_u$ has a unique expression $\iota_+ \alpha + \iota_- \beta$. The factors $1/2$ in SE2 are the explicit coefficients of these maps, not an unrecorded metric change. Define the source-coordinate evaluation map by $\mathcal{E}f_{j,\pm} = q_{j,\pm}$ and matrices $E = (e_1 \ e_2 \ e_3 \ e_4)$, $O = (o_1 \ o_2 \ o_3 \ o_4)$. Then the whole eight-state map is exactly

$$\mathcal{E}(\iota_+ \alpha + \iota_- \beta) = E\alpha + O\beta. \tag{SE3}$$

23 The sign differences are a basis on the literal quartet

The rank of these two pieces can be proved in the original arithmetic quartet, without dropping the phases in a_j . For this purpose form the four polynomials

$$v(r) = (1, -ih'(r), Ah'(r)^2 + 2rh'(r), 7ir^2h'(r) - 13ih'(r)^2)^T.$$

Reduce each polynomial modulo the original h by polynomial division; this changes no evaluation at a root. Write the remainders as $v(r) = V(1, r, r^2, r^3)^T$ modulo h . Their full coefficient matrix is

$$\begin{aligned} V_{1,*} &= (1, 0, 0, 0), \\ V_{2,*} &= (-iC, -2iB, -3i, -4iA), \\ V_{3,*} &= (AC^2 - 9D, 4ABC - 8AD - 7C, \\ &\quad -16A^2D + 4AB^2 - 2AC - 5B, -8A^2C + 4AB - 3), \\ V_{4,*} &= (-13iC^2 + 20iD/A, -52iBC + 76iD + 20iC/A, \\ &\quad 208iAD - 52iB^2 + 5iC + 20iB/A, \\ &\quad 104iAC - 66iB + 20i/A). \end{aligned} \tag{SE4}$$

This is polynomial division by a quartic with its retained leading coefficient A ; the denominators in the last row have not been removed. For the literal u_ρ in GF20, substitution and expansion of the three-by-three determinant remaining after the first row give

$$\det V = 448\delta^2\gamma^2(2\delta^2\gamma^2 - \delta^2 + \gamma^2) > 0. \tag{SE5}$$

The inequality follows since the last factor is $\gamma^2 + \delta^2(2\gamma^2 - 1) > 0$ for $\gamma > 2$. The exact polynomial remainder and determinant calculations in SE4–5 are reproduced by `derive_signed_evaluation.py`; they use rational function arithmetic, not a sampled determinant sign.

Write $\Delta = \det(r_j^n)_{0 \leq n \leq 3, 1 \leq j \leq 4}$ in the same original marking, and retain

$$\chi = A^2 \Delta \prod_{j=1}^4 a_j \in \{1, -1\}. \quad (\text{SE6})$$

Indeed $\prod_j h'(r_j) = A^4 \Delta^2$ by multiplying all pair differences; the six sign reversals have positive product. Squaring SE6 and using $a_j^2 = 1/h'(r_j)$ proves $\chi^2 = 1$. No particular choice of its sign has been imposed. Evaluating SE4 at the roots gives $O = V(r_j^n)_{n,j} \text{diag}(a_1, a_2, a_3, a_4)$, hence

$$\det O = \frac{\chi}{A^2} \det V = 1792 \chi \delta^2 \gamma^2 (2\delta^2 \gamma^2 - \delta^2 + \gamma^2) \neq 0 \quad (\text{SE7})$$

on the literal quartet, where $A = -1/2$ is retained. Thus the four opposite-sign differences are an actual basis in the original source coordinates for every admitted quartet, for either choice of each sign.

The pair means have rank exactly three. Their first coordinate is zero. For a direct lower bound on rank, let E_{pol} be the coefficient matrix of the four polynomials in the first column expression of SE1, again reduced modulo the original h . The minor with rows y, z, w and columns corresponding to r, r^2, r^3 is

$$6i(64A^3D - 16A^2B^2 - 22A^2C + 89AB - 30) = 6i(16\delta^2\gamma^2 - 35\delta^2 + 35\gamma^2) \neq 0. \quad (\text{SE8})$$

The equality follows by substituting GF20. Positivity of the last real factor follows from $16\delta^2\gamma^2 + 35(\gamma^2 - \delta^2) > 0$ under the original bounds $0 < \delta < 1/2, \gamma > 2$. Evaluation at four distinct roots multiplies E_{pol} by the invertible Vandermonde, so its rank and the rank of E are both exactly three.

The remaining even kernel line is also explicit. The full coefficient matrix of the means has rows

$$(0, 0, 0, 0), \quad (0, -1, 0, 0), \quad (3iC, 6iB, 9i, 12iA), \quad (w_0, w_1, w_2, w_3),$$

where

$$\begin{aligned} w_0 &= -2AC^2 + BC + 71D, & w_1 &= -8ABC + 12AD + 2B^2 + 54C, \\ w_2 &= 32A^2D - 8AB^2 + AC + 40B, & w_3 &= 16A^2C - 6AB + 20. \end{aligned} \quad (\text{SE8a})$$

For $W = (r_j^n)_{n,j}$ define

$$t = (3w_3 - 4Aw_2, 0, 4Aw_0 - Cw_3, Cw_2 - 3w_0)^T, \quad \lambda = W^{-1}t. \quad (\text{SE8b})$$

The first coordinate of t is nonzero by SE8. Substitution in the three nonzero rows of SE8a gives zero: the last two equations are the cross-product cancellation between $(C, 3, 4A)$ and (w_0, w_2, w_3) ; the second coordinate of t is zero. Consequently

$$\ker E = \mathbb{C}\lambda, \quad \lambda_j = \frac{t_0(C + Br_j + r_j^2 + Ar_j^3) + t_2(1 + Ar_j) + t_3A}{h'(r_j)}. \quad (\text{SE8c})$$

To verify the last formula, the j th row of W^{-1} is the coefficient row of $h(r)/((r - r_j)h'(r_j))$. Polynomial division gives the four coefficients $h'(r_j)^{-1}(C + Br_j + r_j^2 + Ar_j^3, B + r_j + Ar_j^2, 1 + Ar_j, A)$. Pairing with t , whose r coefficient is zero, gives SE8c. Thus the one even direction lost by evaluation is calculated in the original labelled coordinates, not inferred to be the constant direction from the separate tetrahedral altitude map.

24 The entire eight-state kernel and the original-metric return

Because O is invertible on the literal quartet, SE3 gives the complete kernel, with no unspecified missing sign or root state:

$$\ker \mathcal{E} = \{\iota_+ \alpha - \iota_- O^{-1} E \alpha : \alpha \in \mathbb{C}^4\}, \quad \mathcal{E} \iota_- O^{-1} = I_{\mathbb{C}^4}. \quad (\text{SE9})$$

The inverse here is exactly $\text{adj}(O)/\det O$, with SE1 and SE7 specifying numerator and nonzero denominator. The kernel therefore has dimension four. Its intersection with the even subspace has dimension one by SE8; its intersection with the odd subspace is zero by SE7. These assertions follow by solving $E\alpha + O\beta = 0$; they do not identify the even kernel with a different tetrahedral constant direction by analogy.

The receiving map to the actual conductor target is $\mathcal{E}_W = \Psi \mathcal{E}$, and its source partner is $\mathcal{E}_U = \Phi \mathcal{E}$, with the exact original maps FC32. Thus

$$T_A \mathcal{E}_U = \mathcal{E}_W, \quad \ker \mathcal{E}_U = \ker \mathcal{E}_W = \ker \mathcal{E}, \quad \mathcal{E}_W \iota_- O^{-1} \Psi^{-1} = I_{\mathcal{W}}. \quad (\text{SE10})$$

The last identity constructs a complete section into the sign-odd four-plane of the actual eight-state space. Changing which partner is named $+$ changes the relevant column of O and of ι_- by the same sign; it leaves this section unchanged. Permuting the four labels likewise permutes these two maps inversely, so no root label is implicitly forgotten.

The four original target vectors Ψo_j have the full Gram $O^* G O$, with G from FC56–56a. Consequently

$$\det(O^* G O) = 1792^2 \delta^4 \gamma^4 (2\delta^2 \gamma^2 - \delta^2 + \gamma^2)^2 \det G > 0. \quad (\text{SE11})$$

This retains the entire original Gamma measure and all its cross terms. Their source Gram is $O^* G_U O$; therefore the exact ratio is

$$\frac{\det(O^* G O)}{\det(O^* G_U O)} = \frac{\det G}{\det G_U} = |\det A_{3,s}|^2. \quad (\text{SE12})$$

The first equality cancels the same nonzero explicitly calculated $|\det O|^2$ on both sides, while the second is FC30 with the unchanged K factors. This is a basis constructed from the actual eight signed inverse states and transported through the original linear conductor. Its full sign-odd part is invertible; the rank-three loss in the pair means and the four-dimensional graph kernel SE9 record exactly what the sign-forgetting evaluation removes.

25 An invertible total state map retaining both labels and evaluation

The graph kernel can be resolved by retaining the original root-label vector, rather than discarding the nonlinear evaluation. Define

$$\begin{aligned} \mathcal{J}_U(\iota_+ \alpha + \iota_- \beta) &= (X\alpha, \Phi(E\alpha + O\beta)) \in \mathcal{U} \oplus \mathcal{U}, \\ \mathcal{J}_W(\iota_+ \alpha + \iota_- \beta) &= (Y\alpha, \Psi(E\alpha + O\beta)) \in \mathcal{W} \oplus \mathcal{W}. \end{aligned} \quad (\text{SE13})$$

Both are isomorphisms of the complete eight-dimensional state space. For the target, the entire inverse is

$$\mathcal{J}_W^{-1}(p_1, p_2) = \iota_+ Y^{-1} p_1 + \iota_- O^{-1} (\Psi^{-1} p_2 - E Y^{-1} p_1). \quad (\text{SE14})$$

Substitution in SE13 proves each inverse composition. Replacing Y, Ψ by X, Φ proves the source inverse. Moreover the original conductor itself supplies the exact map between these total state spaces:

$$(T_A|_{\mathcal{U}} \oplus T_A|_{\mathcal{U}}) \mathcal{J}_U = \mathcal{J}_W. \quad (\text{SE15})$$

This is direct substitution of $T_A X = Y$ and $T_A \Phi = \Psi$. In particular, the two original sign states of root j become $(y_j, \Psi q_{j,+})$ and $(y_j, \Psi q_{j,-})$ in the same target direct sum. The root coordinate and the nonlinear input are both retained.

The exact action of signed root permutations on this object includes the terms that evaluation alone would lose. For a permutation π and signs $\sigma_j \in \{1, -1\}$, let $gf_{j,t} = f_{\pi(j), \sigma_j t}$, let $\Pi e_j = e_{\pi(j)}$, and let $Re_j = \sigma_j e_{\pi(j)}$. Thus $g\iota_+ = \iota_+ \Pi$ and $g\iota_- = \iota_- R$. In coordinates (α, z) for the target pair $(Y\alpha, \Psi z)$, SE13–14 give the full transported action

$$\hat{g} = \begin{pmatrix} \Pi & 0 \\ E\Pi - ORO^{-1}E & ORO^{-1} \end{pmatrix}. \quad (\text{SE16})$$

Indeed $z = E\alpha + O\beta$ gives $z' = E\Pi\alpha + OR\beta = ORO^{-1}z + (E\Pi - ORO^{-1}E)\alpha$. The off-diagonal block is therefore evaluated exactly, with all coefficients given in SE1 and SE7. Since SE16 is $\mathcal{J}_W g \mathcal{J}_W^{-1}$ in the displayed coordinates, it obeys the group law for every pair of signed permutations. The actual monodromy subgroup is determined in the signed-root continuation; this formula retains its full action as well. The source uses the identical block matrix in coordinates $(X\alpha, \Phi z)$, so SE15 intertwines every such action.

Every nontrivial even sign change has a nonzero lower-left block, and its rank is exact. Put $F = O^{-1}E$. The image of E is the entire first-coordinate-zero hyperplane by SE8, and the first row of O is (a_1, a_2, a_3, a_4) . Hence $\text{im } F = \{x : \sum_j a_j x_j = 0\}$. Each a_j is nonzero. Projection of this hyperplane onto any proper subset of coordinates is surjective: choose one omitted coordinate to solve the single displayed equation. For a diagonal sign action D_J negating the subset J , the lower-left block of SE16 is $O(I - D_J)F$. Therefore

$$\text{rank}(E - OD_J O^{-1}E) = \begin{cases} 2, & |J| = 2, \\ 3, & |J| = 4. \end{cases} \quad (\text{SE18})$$

For two coordinates this is the just-proved surjective projection; for all four it is twice a rank-three matrix. These are all nonidentity elements of the even-sign kernel. Thus the extra label coordinate in SE13 retains a necessary, quantitatively specified mixing term. SE16 supplies the full action including it.

Put the original orthogonal direct-sum Gamma norms on the two target copies and the original quotient direct-sum norms on the two source copies. In the coordinates just used their Grams are $\text{diag}(G_y, G)$ and $\text{diag}(G_x, G_U)$, respectively. This choice retains each original mass and cross Gram within its copy; it does not identify either $\mathcal{J}$ with an isometry from a previously assigned norm on the abstract finite state space. For $L = T_A|_U$ and every $0 \leq r \leq 8$, the actual inverse-exterior norm is

$$\|\wedge^r (L \oplus L)^{-1}\| = \max_{\substack{p+q=r \\ 0 \leq p, q \leq 4}} \|\wedge^p L^{-1}\| \|\wedge^q L^{-1}\|. \quad (\text{SE17})$$

To prove it, the orthogonal direct sum decomposes its r th exterior space into the orthogonal summands $\wedge^p \mathcal{W} \otimes \wedge^q \mathcal{W}$, $p + q = r$. The map on each summand is the tensor product of the displayed two exterior inverses; in singular bases its norm is the product of their largest singular values. The norm of a block diagonal map on an orthogonal direct sum is the maximum of block norms, proving SE17. Each factor is the original four-plane $A_{3,s}$ inverse appearing in audit (11). This proves the full eight-state return to that bound without deleting the rank-three mean part, the four root labels, either factor sign, or the mixing block.

The next calculation determines the symmetry of the actual cover GF16, with the original ES map and four marked labels retained [1].

26 The actual cover and its eight labelled states

For $u = (u_0, u_2, u_3, u_4)$ put

$$H_u(U, V) = u_0 U^4 + U^3 V + u_2 U^2 V^2 + u_3 U V^3 + u_4 V^4, \quad h_u(x) = H_u(x, 1), \quad A = u_0.$$

The original cubic coefficient remains one throughout. Let

$$\mathcal{B} = \{u : A \operatorname{Disc}(H_u) \neq 0\}, \quad R = \mathbb{C}[u_0, u_2, u_3, u_4, (A \operatorname{Disc} H_u)^{-1}].$$

Over this domain the original source is isomorphic to the finite cover with coordinate algebra

$$S = R[r, a]/(h_u(r), a^2 h'_u(r) - 1). \quad (\text{SM1})$$

In the unchanged original source coordinates its inverse is

$$\begin{aligned} y &= -r - i/a, \\ z &= A/a^3 + 2r/a + 3i/a^2, \\ w &= 7ir^2/a + (u_2 - 17r + Ar^2)/a^2 - 13i/a^3 - 2A/a^4. \end{aligned} \quad (\text{SM2})$$

These formulas are obtained by factoring $H_u = (aU + bV)(cU^3 + dU^2V + eUV^2 + fV^3)$, setting $r = -b/a$, and retaining the resultant equation $C(-b, a) = a^2 h'_u(r) = 1$. Thus SM1 describes the actual inverse states; its two signs are two distinct original source points.

For completeness, $R[r]/(h_u)$ is free of rank four because A is a unit and $A^{-1}h_u$ is monic. The discriminant condition and the Sylvester Bezout identity make $h'_u(r)$ a unit. The second equation of SM1 is therefore the monic relation $a^2 = h'_u(r)^{-1}$, and S is free of rank eight over R . The Jacobian of the two equations with respect to (r, a) has determinant $2ah'_u(r)^2$, a unit. In a sufficiently small target neighbourhood the four simple roots and their two nonzero square roots therefore give eight separate holomorphic inverse charts. This provides the local path-lifting property used below.

Both the base and the actual cover are path connected. The base is the complement in $\mathbb{C}^4$ of the zero set of the nonzero polynomial $A \operatorname{Disc} H_u$. The actual source is the complement of the zero set of its polynomial pullback under the original map P . This pullback is nonzero, since the eight states of any point in $\mathcal{B}$ exist by SM1–2. To connect two points in either complement, restrict its defining polynomial to the complex line through them. Its value at the endpoints is nonzero, so the restricted polynomial is not identically zero and has only finitely many zeros. A path in the complex line parameter from zero to one, with small detours around these finitely many points, connects the endpoints within the complement. This proves the asserted connectedness without an unproved topological classification of the base.

27 The exact upper bound from the retained product invariant

At a base point order the four roots as r_1, r_2, r_3, r_4 and choose one of the two actual values a_j at each root. Denote the eight states by $(j, +)$ and $(j, -)$, corresponding respectively to a_j and $-a_j$ in SM2. During continuation the four roots remain distinct, and their scalings remain nonzero. Put

$$\Delta = \prod_{j < k} (r_k - r_j), \quad \chi = A^2 \Delta \prod_{j=1}^4 a_j.$$

The complete derivative product is

$$\prod_{j=1}^4 h'_u(r_j) = A^4 \prod_{j=1}^4 \prod_{k \neq j} (r_j - r_k) = A^4 \Delta^2.$$

There are six pairs, so the sign from reversing each pair is $(-1)^6 = 1$. Consequently

$$\chi^2 = 1. \quad (\text{SM3})$$

The continuous value χ is therefore constant along every ordered lift of a path. Its initial value may be 1 or -1 ; neither value has been discarded.

A loop in $\mathcal{B}$ induces a root permutation $\pi \in S_4$ and signs $\sigma_j \in \{1, -1\}$ such that $r_j \mapsto r_{\pi(j)}$ and $a_j \mapsto \sigma_j a_{\pi(j)}$. The negative state continues as the negative of the positive one, by uniqueness in the equation $a^2 h'(r) = 1$. The exact action is therefore

$$(j, \epsilon) \mapsto (\pi(j), \sigma_j \epsilon), \quad \epsilon \in \{1, -1\}. \quad (\text{SM4})$$

The final Vandermonde is $\text{sgn}(\pi)\Delta$. Applying the constancy of SM3 gives the necessary relation

$$\prod_{j=1}^4 \sigma_j = \text{sgn}(\pi). \quad (\text{SM5})$$

Thus the monodromy is contained in the explicitly defined group

$$G = \{(\sigma, \pi) \in \{\pm 1\}^4 \rtimes S_4 : \prod_j \sigma_j = \text{sgn}(\pi)\}. \quad (\text{SM6})$$

For each permutation exactly eight sign choices satisfy SM5, so $|G| = 8 \cdot 24 = 192$. As signed permutation matrices its elements have determinant $(\prod_j \sigma_j) \text{sgn}(\pi) = 1$.

28 Explicit local exchanges realizing the full group

Use the real roots

$$(r_T, r_E, r_M, r_N) = (7, 9, 11, 13), \quad A = -1/40.$$

Their sum is forty; hence the cubic coefficient of $A \prod (x - r_j)$ is exactly one. The base target is

$$u_*^{\text{real}} = (-1/40, -59/4, 95, -9009/40) \in \mathcal{B}. \quad (\text{SM7})$$

For the next calculation index the real order as $s_1 = 7, s_2 = 9, s_3 = 11, s_4 = 13$; these indices refer respectively to the unchanged labels T, E, M, N. The four derivatives are

$$(h'(s_1), h'(s_2), h'(s_3), h'(s_4)) = (6/5, -2/5, 2/5, -6/5).$$

Choose the exact base scalings

$$(a_T, a_E, a_M, a_N) = \left(\sqrt{5/6}, -i\sqrt{5/2}, -\sqrt{5/2}, i\sqrt{5/6} \right), \quad (\text{SM8})$$

where the real square roots displayed are positive. Each satisfies the original equation in SM1.

For each adjacent pair (s_j, s_{j+1}) , let $c = (s_j + s_{j+1})/2$ and $d = 1$. Keeping the other two roots fixed, take the explicit path

$$s_j(\theta) = c - de^{i\theta}, \quad s_{j+1}(\theta) = c + de^{i\theta}, \quad 0 \leq \theta \leq \pi. \quad (\text{SM9})$$

The pair sum remains $2c$, so the total root sum and the original coefficient $A = -1/40$ are unchanged. The two moving roots remain distinct. The closed disk of radius one about c contains no other root, so no collision with a fixed root occurs. At the endpoint the pair is interchanged. The unordered roots and all polynomial coefficients consequently return to the same target; SM9 is a loop in the actual base $\mathcal{B}$.

We compute its signs without assuming a braid-group presentation. For a fixed root s_k outside the pair disk, the moving contribution to $h'(s_k)$ is

$$(s_k - s_j(\theta))(s_k - s_{j+1}(\theta)) = (s_k - c)^2 - d^2 e^{2i\theta}.$$

Its values lie in the disk with positive real centre $(s_k - c)^2$ and strictly smaller radius d^2 . This disk is in the right half-plane and has a single continuous logarithm. After the loop that logarithm returns to its initial value. Therefore neither scaling at a fixed root changes sign.

For a moving root, its difference from the other moving root turns by exactly π . Each of its two differences from an outside root lies in a disk not containing zero, with both endpoint values real and of the same sign. A logarithm on the corresponding left or right half-plane shows that the net imaginary part of each of those logarithm differences is zero. The argument change of the complete moving derivative is thus exactly π . Its inverse square root changes phase by $-\pi/2$, with its magnitude taking the new root's required magnitude. By the explicit choices SM8, the resulting permutation is the four-cycle

$$\beta_j = (j + \quad (j + 1) + \quad j - \quad (j + 1) -). \quad (\text{SM10})$$

Equivalently, in the four retained Time labels the three generators are

$$\begin{aligned} \beta_1 &= (\text{T} + \text{E} + \text{T} - \text{E} -), \\ \beta_2 &= (\text{E} + \text{M} + \text{E} - \text{M} -), \\ \beta_3 &= (\text{M} + \text{N} + \text{M} - \text{N} -). \end{aligned} \quad (\text{SM11})$$

Every sign and every fixed state has been determined by the path.

The square β_j^2 flips exactly the two adjacent root signs. These three pair flips generate all even sign patterns: as vectors over $\mathbb{F}_2$ they are $(1, 1, 0, 0)$, $(0, 1, 1, 0)$ and $(0, 0, 1, 1)$, which are independent and span the dimension-three subspace with coordinate sum zero. The projections of the three β_j are the three adjacent transpositions, and they generate all of S_4 by successively moving entries to any prescribed positions. Thus the group generated by these actual loops contains all eight elements in the sign kernel and projects onto all 24 permutations. It has at least $8 \cdot 24 = 192$ elements. The universal upper bound SM6 has exactly 192 elements, proving equality:

$$\boxed{\text{Mon}(P^{-1}(\mathcal{B})/\mathcal{B}) = G.} \quad (\text{SM12})$$

This proof establishes the entire group without an unproved claim that every base loop is a braid word.

29 Transport from the literal original quartet

Start with the original marking

$$(\rho_M, \rho_N, \rho_T, \rho_E) = \left(\frac{1}{2} + \delta + i\gamma, \frac{1}{2} + \delta - i\gamma, \frac{1}{2} - \delta + i\gamma, \frac{1}{2} - \delta - i\gamma \right), \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2.$$

Its target is $H_\rho = -\frac{1}{2} \prod (U - \rho_j V)$, with the original cubic coefficient one. The following two paths give a fully explicit base transport to SM7. First, for $0 \leq t \leq 1$, put

$$\begin{aligned} r_M(t) &= \frac{1}{2} + (1-t)(\delta + i\gamma) + t, & r_N(t) &= \frac{1}{2} + (1-t)(\delta - i\gamma) + 3t, \\ r_T(t) &= \frac{1}{2} + (1-t)(-\delta + i\gamma) - 3t, & r_E(t) &= \frac{1}{2} + (1-t)(-\delta - i\gamma) - t. \end{aligned} \quad (\text{SM13})$$

Their sum is always two, so $A = -1/2$ throughout this segment. Roots with different original imaginary signs have different imaginary parts for $t < 1$. Roots with the same original imaginary sign have real difference $2\delta(1-t) + 4t$, which is positive. At $t = 1$ the four real roots are $(3/2, 7/2, -5/2, -1/2)$ and are distinct. Thus the entire segment remains in $\mathcal{B}$.

Next translate all four roots by $19t/2$, $0 \leq t \leq 1$, and put $A(t) = -1/(2 + 38t)$. Their differences remain fixed, their sum is $2 + 38t \neq 0$, and the retained original cubic coefficient is again one. At its endpoint the marked roots are $(11, 13, 7, 9)$, namely SM7 in the original Time order. This second segment also lies entirely in $\mathcal{B}$.

The actual signs transport by the original equations, not by discarding a branch. An exact formula along either differentiable segment is

$$a_j(t) = a_j(0) \exp \left[-\frac{1}{2} \int_0^t \left(\frac{A'(s)}{A(s)} + \sum_{k \neq j} \frac{r'_j(s) - r'_k(s)}{r_j(s) - r_k(s)} \right) ds \right]. \quad (\text{SM14})$$

The integrand has no singularity on these paths. Differentiating shows that

$$a_j(t)^2 A(t) \prod_{k \neq j} (r_j(t) - r_k(t))$$

is constant, and its initial value is one. Hence SM14 is precisely the lifted inverse branch. Both initial signs lift separately. Using SM8 as the signs at the real endpoint defines their exact transported signs at the original quartet; any other initial sign convention conjugates SM11 by a diagonal sign permutation. It does not change SM12, because the determinant-one subgroup is invariant under that conjugation.

Conjugating each loop SM9 by the explicitly constructed path SM13 and its translation segment gives based loops at the literal quartet with the same marked-root action. This transport concerns the auxiliary quartic targets of the fixed nonlinear map. It does not assert that intermediate roots are arithmetic zeta zeros or that an original conductor period is moved along these paths.

30 Why this group is not the Weyl group of type D_4

With composition acting first by the rightmost generator, the product $\beta_1 \beta_2 \beta_3$ is the eight-cycle

$$(T + E + M + N + T - E - M - N -). \quad (\text{SM15})$$

This follows by applying the three four-cycles SM11 to each state in order. It has order eight.

For comparison, the reflections in the roots $\{\pm e_j \pm e_k : j \neq k\}$ generate precisely the signed permutations with an even total number of sign changes. Indeed the reflection in $e_j - e_k$ swaps those coordinates, and the reflection in $e_j + e_k$ swaps them with both signs changed. Their compositions generate every permutation and every pair sign flip, and each such reflection has even total sign product. This proves the usual signed-permutation description of the Weyl group $W(D_4)$ directly:

$$W(D_4) = \{(\sigma, \pi) : \prod_j \sigma_j = 1\}.$$

This group has no element of order eight. An underlying permutation of four coordinates can have cycle lengths only 1, 2, 3, 4. A signed cycle of length m has order m if its sign product is positive and $2m$ if its sign product is negative, as follows by applying the cycle m times to each coordinate. With no length-four cycle, the available orders combine to divisors of four or six. With a length-four cycle, the even total sign condition makes its sign product positive, so its order is four. Thus order eight never occurs in $W(D_4)$, whereas SM15 occurs in G . The groups are therefore not even abstractly isomorphic, despite their equal order 192.

31 Exact deck groups

Let τ denote simultaneous reversal of all four root signs, $\tau(j, \epsilon) = (j, -\epsilon)$. The centralizer of G in the full permutation group of the eight inverse states is

$$C_{S_8}(G) = \{1, \tau\}. \quad (\text{SM16})$$

Here is a complete proof. Fix one state, say $(1, +)$. Its stabilizer in G also fixes $(1, -)$. No state at another root is fixed by that stabilizer: a pair sign flip involving that root and one of the other two roots different from root 1 belongs to the stabilizer and moves that state. Hence the stabilizer has exactly the two fixed states $(1, +), (1, -)$. Any permutation c commuting with G must send $(1, +)$ to one of these fixed states. The group G is transitive on all eight states (permutations move root labels and pair flips change either chosen sign), so the image of this one state determines c everywhere through $c(gx) = gc(x)$. The first choice gives the identity and the second gives τ . Both commute with every signed permutation, proving SM16.

A deck transformation of the actual connected cover restricts on the base fibre to a permutation commuting with monodromy: apply the deck transformation to a lifted loop and use uniqueness of path lifting in the local charts of SM1. Its restriction is injective as a function of the deck transformation. Indeed a transformation fixing one fibre point fixes the lift of every path starting there; connectedness, or the explicit path-lifting description, reaches the entire cover. Thus SM16 is an upper bound of two for the actual deck group.

Both transformations are realized. The nontrivial one is the original rational factor-sign involution, in unchanged source coordinates,

$$\Sigma(a, y, z, w) = \left(-a, y + \frac{2i}{a}, \frac{6i}{a^2} - z, w - \frac{14iy^2}{a} + \frac{28y}{a^2} + \frac{40i}{a^3} \right). \quad (\text{SM17})$$

To check it directly in the inverse model, keep the same r, u in SM2 and replace a by $-a$. Its new y is $y + 2i/a$ and its new z is $6i/a^2 - z$. Expanding the two expressions for w gives exactly SM17. It follows from this same operation that $\Sigma^2 = 1$ and $P\Sigma = P$ on $a \neq 0$. Since $u_0 = ac$ is nonzero on $\mathcal{B}$, every point of its preimage has $a \neq 0$. SM17 is regular on the entire cover and realizes τ . The exact cover deck group is consequently

$$\text{Aut}_{\mathcal{B}}(P^{-1}(\mathcal{B})) = \{1, \Sigma\} \cong C_2. \quad (\text{SM18})$$

The same two elements are all rational birational deck transformations of the original polynomial map. To justify the passage from rational maps, write $K_0 = \mathbb{C}(u_0, u_2, u_3, u_4)$ and $L_0 = \mathbb{C}(a, y, z, w)$ with the embedding defined by P . SM1 makes $[L_0 : K_0] = 8$. The actual ring S is also the localization

$$\mathbb{C}[a, y, z, w, (P_0 \text{ Disc}(H_{P(q)}))^{-1}],$$

so it is integrally closed: a localization of a polynomial unique-factorization ring has this property, since a reduced fraction integral over it has denominator dividing a power of its relatively prime numerator and therefore has unit denominator. Since S is finite integral over R , it is the full integral closure of R in L_0 . In fact any element of L_0 integral over R is integral over S by the same monic polynomial, and integrally closedness puts it in S . Every K_0 -automorphism of L_0 preserves this integral closure, and hence extends to a regular deck transformation on the actual cover. SM18 therefore bounds it by those same two elements; SM17 realizes both. Thus

$$\text{Aut}_{K_0}(L_0) = \{1, \Sigma\}. \quad (\text{SM19})$$

This statement is about birational automorphisms fixing the target field, not arbitrary rational maps of the source.

Finally the only global polynomial deck automorphism D of $\mathbb{C}^4$ satisfying $P \circ D = P$ is the identity. Such an automorphism would restrict to SM19. The nonidentity element cannot

extend polynomially across $a = 0$, because its second coordinate is $(ay + 2i)/a$, whose numerator is not divisible by a . Equality on the dense open set $a \neq 0$ forces equality as rational functions, so there is no alternative polynomial extension. This proves the assertion for the whole original source, while retaining the nontrivial regular deck transformation on the precise eight-sheet domain. This does not classify polynomial automorphisms satisfying the different equation $P \circ D = D \circ P$; that centralizer is not asserted to be trivial.

32 Receiving the group in the fixed original conductor target

For the fixed original receiving isomorphism $\Psi : \mathbb{C}^4 \rightarrow W = \mathcal{P}_3$, with all original moments and Gamma norms retained, set $P_W = \Psi P \Psi^{-1}$. Its eight-sheet base is exactly $\Psi(\mathcal{B})$. Sending a target path $u(t)$ to $\Psi u(t)$ and each source state to its image under Ψ gives a bijection of all path lifts, with inverse Ψ^{-1} . Thus its monodromy is exactly SM12 on the eight transported marked states, and its nontrivial rational deck transformation is exactly $\Psi \Sigma \Psi^{-1}$. The cover deck group is C_2 and its global polynomial deck group is trivial by the same conjugacy.

These are statements about the constructed nonlinear cover on the original target. They do not enlarge the symmetry group of a fixed Segre pairing or assert that the original conductor itself is a nonlinear map. The original linear conductor remains the exact invertible intertwiner proved in FC26–35 above. All four Time labels and both signs over each are present in the monodromy action just calculated.

References

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- [2] The Clankers, *Arithmetic Shells, Finite Support Geometry, and Lorentz Lifts: Exact Identities, Certificates, and Symmetry Constructions*, dated 4 August 2026, `es_fable_c123_preprint.tex`, consulted source snapshot, 19–20 September 2026. Exact theorem/equation labels are cited at use. Source SHA-256: 0e71a9c146eb860a9b66a2287dca673ab675e894e545df0ede642e74a4f353c4. The four-Time and conductor comparison in this note is a new calculation, not an attribution to the human authors of the original Jacobian example.
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- [6] Terence Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, <https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/>. Equations (1) and the displayed three-point fibre record the Alpöge–Fable example. The ES four-Time locale is an additional marked construction; it must not be credited as a fourth preimage in that three-dimensional fibre.